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Social Attention as an Exogenous Signal for Short-Horizon Market Volatility Forecasting

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Abstract

This thesis develops and evaluates a sentiment classifier for StockTwits posts related to the SPY exchange-traded fund. The analysis progresses from an off-the-shelf zero-shot FinBERT model to a supervised fine-tuned classifier, and then evaluates two semi-supervised approaches under the same experimental protocol: self-training with confidence- and neighbourhood-based pseudo-label filtering, and graph-based label spreading. While both semi-supervised methods achieve a modest improvement over supervised fine-tuning, the gains are too small to justify their additional complexity. The fine-tuned supervised model is therefore retained for the forecasting stage.

The resulting sentiment signal is summarised through three complementary dimensions—level, dispersion, and message volume—and combined with financial variables, including the CBOE Volatility Index (VIX), to forecast returns, market direction, and realised volatility across multiple horizons. The results show that financial social media sentiment does not provide a reliable directional trading signal: no configuration consistently outperforms a naive benchmark for returns, while financial variables remain the strongest predictors of market direction beyond the shortest horizon. In contrast, sentiment information improves short-term realised volatility forecasting even after accounting for the VIX. This improvement is driven primarily by message volume, a proxy for retail investor attention, rather than by sentiment polarity or dispersion, and is concentrated in specific market regimes. Overall, the findings suggest that retail attention extracted from sentiment-labelled posts contains useful regime-dependent information for short-term volatility forecasting, but does not provide a robust signal for predicting market direction.

Keywords: Sentiment Analysis, FinBERT, Semi-Supervised Learning, Financial Forecasting, StockTwits, Volatility Forecasting, Retail Attention

Abstract (Italian Version)

Questa tesi sviluppa e valuta un classificatore di sentiment per i post di StockTwits relativi all'ETF SPY. L'analisi procede a partire da un modello FinBERT zero-shot pre-addestrato, per poi passare al fine-tuning supervisionato su dati etichettati e, infine, al confronto di due approcci semi-supervisionati valutati secondo lo stesso protocollo sperimentale: self-training con pseudo-etichette filtrate sulla base della confidenza e del vicinato, e label spreading basato su grafi. Sebbene entrambi gli approcci semi-supervisionati producano un lieve miglioramento rispetto al modello supervisionato, il beneficio ottenuto non giustifica la maggiore complessità introdotta. Di conseguenza, il modello supervisionato fine-tuned viene adottato per la fase di previsione.

Il segnale di sentiment ottenuto viene quindi sintetizzato attraverso tre dimensioni complementari — livello, dispersione e volume dei messaggi — e combinato con variabili finanziarie, tra cui il CBOE Volatility Index (VIX), per prevedere rendimenti, direzione del mercato e volatilità realizzata su diversi orizzonti temporali. I risultati mostrano che il sentiment estratto dai social media finanziari non costituisce un segnale affidabile per la previsione della direzione del mercato: nessuna delle configurazioni analizzate supera in modo consistente un benchmark ingenuo nella previsione dei rendimenti, mentre le sole variabili finanziarie rimangono i migliori predittori della direzione oltre l'orizzonte temporale più breve. Al contrario, il sentiment migliora la previsione della volatilità realizzata nel breve termine anche dopo aver considerato il VIX. Tale miglioramento è trainato principalmente dal volume dei messaggi, interpretabile come una misura dell'attenzione degli investitori retail, piuttosto che dalla polarità o dalla dispersione del sentiment, ed è concentrato in specifici regimi di mercato. Nel complesso, i risultati mostrano che l'attenzione degli investitori retail estratta da post etichettati tramite analisi del sentiment contiene informazioni utili, dipendenti dal regime di mercato, per la previsione della volatilità di breve termine, ma non fornisce un segnale robusto per prevedere la direzione del mercato.

Parole chiave: Analisi del Sentiment, FinBERT, Apprendimento Semi-Supervisionato, Previsione Finanziaria, StockTwits, Previsione della Volatilità, Attenzione degli Investitori Retail

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1 | Introduction

Financial markets are driven not only by quantitative factors such as prices, trading volumes, and macroeconomic indicators, but also by qualitative information conveyed through news articles, corporate disclosures, social media, and analyst reports. This textual information often carries signals about investor expectations, uncertainty, and overall market mood, that may precede movements in observable market variables.

Recent advances in Natural Language Processing (NLP) have made it possible to analyse such textual information systematically and at scale. In particular, sentiment analysis has attracted growing interest as a tool for capturing the tone embedded in financial text and translating it into quantitative signals that can be integrated into predictive models.

Despite these developments, a significant obstacle remains: the limited availability of labelled financial text. Most state-of-the-art sentiment models rely on supervised learning and therefore require large, high-quality annotated datasets. In the financial domain, however, producing such annotations is not only costly and time-consuming but also intrinsically difficult. Financial language is often technical, context-dependent, and perhaps carries ambiguity: the same expression may convey opposite implications depending on the surrounding context, and tone frequently rests on subtle cues rather than explicit wording. As a result, manual labelling tends to be subjective and prone to inconsistency, which further limits the size and reliability of the labelled corpora available. The combined effect is that vast amounts of potentially valuable unlabelled text remain unexploited.

This thesis investigates whether sentiment extracted from financial social media text adds genuine predictive value for financial forecasting, and specifically whether it continues to do so once measured against a financial baseline that already includes the market's own volatility benchmark, the CBOE Volatility Index (VIX). Answering this question convincingly first requires a sentiment signal that can be trusted, rather than one accepted at face value: this thesis therefore also examines, as a necessary preliminary step, whether semi-supervised learning – which combines a limited set of labelled examples with a much larger pool of unlabelled financial text – can improve on a purely supervised sentiment classifier under the labelling constraints just described.

The thesis is accordingly organised in two parts. The first develops and validates a

sentiment classifier for StockTwits posts about the SPY exchange-traded fund, progressing from an off-the-shelf, zero-shot model to a fine-tuned classifier, and then testing two complementary semi-supervised strategies – self-training and graph-based label spreading – against that fine-tuned baseline under a matched evaluation protocol. The second part uses the resulting sentiment signal, decomposed into level, dispersion, and volume-based features, to test directly whether it improves the forecasting of returns, direction, and volatility over multiple horizons, once compared against a financial baseline that already includes the VIX.

This structure yields two distinct empirical contributions. The first is a systematic, like-for-like comparison of two semi-supervised strategies for financial sentiment classification, including a direct validation of pseudo-label quality against withheld ground truth – a comparison that, to the best of this thesis’s knowledge, has not previously been carried out for financial social media text under a matched embedding space. The second, and central, contribution is an assessment of whether the resulting sentiment signal survives comparison against the toughest available financial baseline, and, if so, which specific aspect of that signal – its level, its dispersion, or the volume of discussion it reflects – is responsible.

The remainder of the thesis is organised as follows. Chapter 2 presents the objectives and research questions. Chapter 3 reviews the main NLP methods and the literature on sentiment-based financial forecasting. Chapter 4 describes the dataset and methodology for sentiment analysis. Chapter 5 presents the predictive models and empirical results. Chapter 6 concludes and outlines possible future developments.

2 | Objectives

The increasing availability of textual financial information—including news articles, corporate disclosures, and market commentaries—has stimulated growing interest in the use of sentiment analysis for financial forecasting. While recent advances in NLP have substantially improved the ability to extract sentiment from text, most existing approaches rely on supervised learning techniques that require large amounts of labelled data. In financial domains, however, obtaining high-quality labelled datasets is often costly and time-consuming, leaving a substantial amount of potentially valuable unlabelled information unused.

2.1. Research Gap

Although sentiment analysis is increasingly applied to financial forecasting, the existing literature presents two recurring limitations. First, most sentiment-extraction methods are supervised and depend heavily on the availability of labelled financial corpora, which restricts their applicability when annotated data are scarce. Second, studies that address data scarcity rarely assess whether the resulting sentiment indicators add genuine predictive value once traditional financial and market variables are already accounted for. The potential of semi-supervised learning to exploit large volumes of unlabelled financial text, together with the marginal contribution of the resulting indicators to forecasting performance, therefore remains only partially explored. This thesis aims to address this gap.

2.2. Objectives

Against this background, the main objective of this thesis is twofold. The first is to investigate whether semi-supervised learning can enhance the sentiment-extraction process by effectively leveraging large volumes of unlabelled financial text, thereby improving the quality and robustness of the resulting sentiment indicators. The second is to assess whether these indicators, once integrated into a forecasting framework, can improve the prediction of market-related variables beyond what is achievable with conventional

quantitative indicators alone.

To this end, the study integrates the sentiment indicators within a forecasting framework that also incorporates traditional historical and financial variables. By comparing predictive models with and without sentiment-related information, the research evaluates the extent to which sentiment contributes additional explanatory and predictive power. More broadly, the thesis seeks to provide empirical evidence on the practical benefits and limitations of semi-supervised sentiment analysis in real-world financial forecasting applications.

2.3. Research Questions

The research is guided by four main questions:

- RQ1.** Can sentiment extracted from financial textual data provide meaningful information for predicting market-related outcomes?
- RQ2.** Can semi-supervised learning techniques improve sentiment-analysis performance when only a limited amount of labelled financial data is available?
- RQ3.** Does the inclusion of sentiment-based indicators enhance forecasting accuracy beyond what can be achieved using traditional financial and market variables alone?
- RQ4.** What are the main challenges, limitations, and opportunities associated with the application of semi-supervised sentiment analysis in financial forecasting contexts?

3 | Literature Review

This chapter reviews the literature that motivates and grounds the methodological choices made in the rest of this thesis. The material is organised around the two parts into which the thesis itself is divided. Sections 3.1–3.3 cover the first part, concerned with obtaining a reliable sentiment signal from financial social media text: they move from the general mechanics of representing text numerically (Section 3.1), to the specific problem of adapting sentiment analysis to financial language (Section 3.2), to the semi-supervised learning techniques used to extend a small labelled seed set when reliable financial sentiment labels are scarce (Section 3.3). Sections 3.4 and 3.5 cover the second part, concerned with whether that sentiment signal is actually useful for forecasting: they review the empirical evidence on whether market sentiment, disagreement, and attention relate to prices, volatility, and trading activity (Section 3.4), and the deep learning approaches used to combine textual and numerical features for financial prediction, with particular attention to volatility forecasting and the VIX (Section 3.5). The chapter closes with a summary (Section 3.6) that draws these two halves together and states the specific gaps this thesis is positioned to address.

3.1. Natural Language Processing Techniques

The first step regarding sentiment analysis is to transform text into a representation that a machine can process. In this sense, words are converted into numerical objects, typically vectors, that represent semantic or contextual information. Once text has been transformed into a suitable representation, machine learning models may use it for downstream tasks such as classification, regression, or forecasting.

3.1.1. Tokenisation

Tokenisation is the process of splitting text into smaller units, commonly called tokens. Traditional tokenisation methods rely on spaces and punctuation, whereas more modern approaches often use subword tokenisation in order to handle rare words and reduce vocabulary size.

Among the most widely used subword techniques are Byte Pair Encoding (BPE) (Sennrich, Haddow, and Birch, 2016) and WordPiece, the scheme used by BERT-family models (Devlin et al., 2019). These methods are particularly relevant in transformer-based models, where they help balance vocabulary efficiency and semantic coverage, and they carry a practical advantage for financial text specifically: they allow a model to represent rare or previously unseen tokens, such as stock tickers, cashtags, and finance-specific abbreviations, as a combination of smaller known subword units rather than mapping them to a single generic “unknown” token. This is particularly relevant for the informal, abbreviated register typical of financial social media, where such tokens are common and a fixed, closed-vocabulary tokeniser would otherwise discard useful information.

3.1.2. Text Encoding

After tokenisation, text must be encoded numerically. Early approaches used one-hot representations or bag-of-words models, which represent a document as an unordered collection of token counts or indicators; both ignore word order and treat every pair of distinct tokens as equally unrelated, regardless of their meaning. More advanced methods introduced dense vector representations, known as word embeddings, which place semantically related words close together in a continuous vector space. Mikolov et al. (2013) introduce Word2Vec, which learns such embeddings by predicting a word from its context or vice versa, and Pennington, Socher, and Manning (2014) introduce GloVe, which instead derives embeddings from global word co-occurrence statistics computed over an entire corpus. Both produce a single, fixed vector per word regardless of the sentence in which it appears, which is a major limitation for words whose meaning is context-dependent – a problem that is especially acute in financial language, where the same word may carry different, sometimes opposite, implications depending on context (a point revisited in Section 3.2).

Peters et al. (2018) tackle this limitation with ELMo, which computes a word’s representation as a function of the entire sentence using a bidirectional recurrent language model, so that the same word receives different embeddings in different contexts. This idea of contextual embeddings was then taken further, and made substantially more effective and parallelisable, by the transformer architecture described next.

3.1.3. Transformer Architecture

The transformer architecture, introduced by Vaswani et al. (2017), has become the dominant framework in modern NLP. Unlike recurrent models, which process a sequence one token at a time and propagate information step by step, transformers rely on self-attention

components that let every token attend directly to every other token in a sequence in a single step. This allows transformers to model extended dependencies more effectively and to process an entire sequence in parallel during training, substantially reducing training time compared to sequential architectures.

Models such as BERT (Devlin et al., 2019), RoBERTa (Liu, Ott, Goyal, et al., 2019), and FinBERT (Araci, 2019) are built on this architecture and are widely used for financial sentiment analysis. Their main advantage lies in the combination of two properties that earlier embedding methods offered only separately: the contextual sensitivity introduced by models such as ELMo, and the ability to adapt an already pretrained model to a specialised domain through fine-tuning on a comparatively small amount of task- or domain-specific labelled data, as discussed further in Section 3.2.

This chapter reviews the literature that motivates and grounds the methodological choices made in the rest of this thesis. The material is organised around the two parts into which the thesis itself is divided. Sections 3.1–3.3 cover the first part, concerned with obtaining a reliable sentiment signal from financial social media text: they move from the general mechanics of representing text numerically (Section 3.1), to the specific problem of adapting sentiment analysis to financial language (Section 3.2), to the semi-supervised learning techniques used to extend a small labelled seed set when reliable financial sentiment labels are scarce (Section 3.3). Sections 3.4 and 3.5 cover the second part, concerned with whether that sentiment signal is actually useful for forecasting: they review the empirical evidence on whether market sentiment, disagreement, and attention relate to prices, volatility, and trading activity (Section 3.4), and the deep learning approaches used to combine textual and numerical features for financial forecasting, with particular attention to volatility forecasting and the VIX (Section 3.5). The chapter closes with a summary (Section 3.6) that draws these two halves together and states the specific gaps this thesis is positioned to address.

3.2. Sentiment Analysis

Sentiment analysis, also referred to as opinion mining, is the task of identifying and extracting subjective information from text – typically the polarity of an opinion (positive, negative, or neutral), but more generally also its holder, its target, and its intensity. Pang and L. Lee (2008) survey the field’s early development and distinguish two broad methodological families that recur throughout its history. Lexicon-based methods score a document by matching its words against a precompiled list of polarity-bearing terms and aggregating the result; they require no labelled training data and are easy to interpret, but they struggle with negation, sarcasm, and domain-specific vocabulary, since a fixed

word list cannot adapt its polarity assignments to context. Supervised methods instead treat sentiment classification as an ordinary text classification problem, training a model on documents labelled with their true polarity; they capture context and domain-specific usage far better, but their accuracy is bounded by the quantity and quality of labelled data available for training – the same constraint that motivates the semi-supervised methods reviewed in Section 3.3.

This trade-off is sharpened considerably once the domain shifts to finance. Financial text uses ordinary words in ways that general-purpose sentiment resources do not anticipate: a term such as “liability,” “tax,” or “cost” carries a negative connotation in everyday language but is often descriptively neutral in a regulatory filing, while other words acquire a specifically financial and clearly polarised meaning that a general lexicon does not capture at all. Loughran and McDonald (2011) demonstrate this concretely by applying a widely used general-purpose psychological dictionary to a large sample of 10-K filings and finding that roughly three-quarters of the words it flags as negative are not actually negative in a financial reporting context; they address the problem by constructing a finance-specific dictionary directly from financial disclosures rather than adapting a general-purpose one. Malo et al. (2014) take the supervised route to the same domain problem, introducing the Financial PhraseBank, a corpus of sentences drawn from financial news and annotated for polarity by finance-literate annotators, which has since become a standard benchmark for training and evaluating financial sentiment classifiers directly on in-domain, human-labelled text rather than through a hand-built word list.

The transformer-based language models introduced in Section 3.1.3 offer a third route to the same adaptation problem: instead of building a domain-specific dictionary or training a classifier from scratch, a general-purpose pretrained model can be fine-tuned on a modest amount of in-domain labelled text. Araci (2019) apply this idea to finance directly, fine-tuning BERT on a large unlabelled financial corpus before further fine-tuning it on the Financial PhraseBank, and show that the resulting model, FinBERT, outperforms both dictionary-based methods and general-purpose sentiment classifiers on financial sentiment classification. FinBERT has since become close to a default choice for financial sentiment tasks, including the classifier developed in this thesis, precisely because it combines the contextual sensitivity of a transformer with explicit adaptation to financial vocabulary.

This progression – from hand-built dictionaries, to supervised classifiers trained on curated financial benchmarks, to fine-tuned domain-adapted transformers – mirrors the general trajectory of NLP described in Section 3.1, but it does not fully resolve the problem once the object of study shifts from news articles and regulatory filings to financial social media. Resources such as the Financial PhraseBank consist of short, well-formed,

professionally written sentences; social media posts, by contrast, are informal, frequently abbreviated, and idiosyncratic, which makes them harder to label consistently even for a finance-literate annotator. Labelled data of this specific kind – financial social media posts annotated with a reliable sentiment label – is scarcer and more expensive to produce than either of the two resources discussed above, precisely because it requires annotators to be simultaneously finance-literate and familiar with the conventions of the platform in question. It is this specific scarcity that motivates the semi-supervised learning strategies reviewed next.

3.3. Semi-Supervised Learning

In a wide range of real-world applications, the acquisition of large volumes of labelled data proves both costly and time-consuming, while unlabelled data is available in considerably greater quantities. Semi-supervised learning (SSL) addresses this limitation by proposing a unified training framework that jointly exploits a small labelled set together with a larger unlabelled subset. Formally, the available data can be divided into a labelled set $\mathcal{X} = \{(x_i, y_i)\}_{i=1}^n$ and an unlabelled set $\mathcal{U} = \{u_j\}_{j=1}^m$, where usually $m \gg n$. The objective is to improve generalisation by using not only the explicit label information contained in $\mathcal{X}$, but also the distributional and geometric structure of $\mathcal{U}$.

Most SSL methods rest on two related assumptions: the *smoothness assumption*, whereby observations that are close in the feature or embedding space are likely to share similar labels, and the *cluster assumption*, whereby decision boundaries should preferably lie in low-density regions of the data distribution. Together, these assumptions justify the use of unlabelled data: even without label information, $\mathcal{U}$ reveals the geometric and distributional structure of the input space, which can be exploited to constrain the learning process. This is especially relevant in modern NLP, where texts are mapped into dense embedding spaces produced by pre-trained language models, and where the geometry of that space is informative about semantic and, potentially, sentiment similarity.

Within SSL, several methodological families operationalise these assumptions differently, including self-training, consistency regularisation, and graph-based label propagation. This thesis relies on two of them, chosen for their complementary logic: self-training, which is model-driven and expands the training set through the classifier’s own predictions, and label spreading, which is structure-driven and propagates label information through a similarity graph. The remainder of this section reviews each family in turn, situates it within prior applications to sentiment classification, and closes with a synthesis that motivates comparing the two in the financial sentiment setting addressed by this thesis.

3.3.1. Self-Training and Pseudo-Labelling

Self-training is a semi-supervised paradigm in which a model trained on labelled data is used to generate labels, called *pseudo-labels*, for unlabelled observations; these pseudo-labels are then treated as additional training targets in subsequent iterations. In its simplest form, a base model is trained on $\mathcal{X}$, applied to the samples in $\mathcal{U}$, and assigns to each of them the class with the highest predicted probability. The pseudo-labelled samples are then added to the training set and the model is retrained or fine-tuned on the enlarged corpus. The appeal of the approach lies in its simplicity: it turns unlabelled data into usable training examples without requiring a change of model architecture, which is particularly attractive when manual labelling is expensive, subjective, or slow to obtain – as is the case for large volumes of market-related social media posts, where assigning a reliable Bullish or Bearish label to every observation is costly.

Self-training predates the recent wave of transformer-based NLP and has a specific history in sentiment classification. He and Deyu Zhou (2011) bootstrap a sentiment classifier from a small set of labelled polarity-bearing features rather than labelled documents, iteratively enlarging the training set with self-labelled instances above a confidence threshold. S. Lee and Kim (2017) extend this idea with a dedicated sentiment-labelling module that filters and injects only the most reliable pseudo-labelled reviews into the training corpus, reporting consistent gains over classifiers trained on the seed set alone. Sazzed and Jayarathna (2021) combine a lexicon-based analyser with a machine-learning classifier in a self-supervised pipeline (SSentiA), using the confidence scores produced by the lexicon side to generate pseudo-labels for the supervised side, while Jafarlou and Kubek (2024) revisit confidence-based self-training explicitly as a way of reducing annotation costs on modern sentiment benchmarks. Across these works, spanning more than a decade of research, the underlying trade-off has not changed: enlarging the training set with pseudo-labels only helps if the pseudo-labels are trustworthy, and trustworthiness must be estimated without access to the true label.

Confidence-Based Filtering and Confirmation Bias

The most direct way to estimate trustworthiness is confidence-based filtering: only pseudo-labels whose predicted probability exceeds a fixed threshold are retained, while the rest are discarded or left unlabelled. The strategy is intuitive, but a fixed threshold is a blunt instrument. If it is set too high, especially early in training when the model is still uncertain, most unlabelled data are rejected and the model continues to rely almost exclusively on the small labelled set, gaining little from the unlabelled pool. If it is set too low, noisy or incorrect pseudo-labels enter the training set; because the model is trained

on its own predictions, these errors tend to reinforce themselves over successive iterations rather than being corrected, a failure mode generally known as *confirmation bias*. The problem is compounded by the fact that neural networks can be confidently wrong, which is particularly likely in noisy, informal, or context-dependent text such as social media posts. L. Zhu, Ke, and Lau (2023a) show that a constant high threshold can cause a model to reject nearly all pseudo-labels at the start of training, leaving the decision boundary effectively dominated by the labelled samples alone, and propose replacing the fixed cutoff with a filter that adapts to the evolving confidence distribution of the model. This line of work illustrates a broader shift in the SSL literature away from a single hand-tuned threshold and towards filtering criteria that are sensitive to how reliable a given confidence score actually is at a given point in training.

Neighbourhood- and Distance-Based Selection

A complementary shift addresses a different weakness of confidence alone: a prediction can be confident yet isolated, inconsistent with nearby examples in the embedding space, or systematically biased towards classes that are easier to predict. Distance-based selection strategies exploit the geometry of the representation space to correct for this: since pre-trained language models map text into dense embeddings, an unlabelled sample whose neighbours mostly agree with its predicted class can be considered more reliable than one sitting in a mixed or contradictory neighbourhood, independently of the raw confidence score. Botzer et al. (2023) operationalise this idea in TK-KNN, a pseudo-labelling strategy for intent classification that scores unlabelled samples using both model confidence and similarity to labelled examples in embedding space, and additionally enforces a balanced, per-class top- k selection instead of a global threshold; the authors show that confidence-only selection tends to produce pseudo-label sets skewed towards easier classes, amplifying class imbalance over successive iterations. Sun et al. (2023a) apply a related idea to noisy distantly-supervised relation extraction, using a weighted k -nearest-neighbour graph to identify which labelled instances are reliable enough to keep, and treating the rest as unlabelled rather than discarding them outright – a reminder that uncertain observations can still be informative if handled carefully, rather than being a pure source of noise to be removed. Although neither of these two works addresses sentiment classification directly, they converge with the confirmation-bias literature above on the same conclusion: a pseudo-label is more trustworthy when confidence and local consistency in embedding space agree, and selection strategies that consider only one of the two are more exposed to noise and imbalance.

This body of work is directly relevant to the present setting. Financial social media

posts are frequently short, informal, and ambiguous in isolation, which makes model confidence, on its own, an unreliable filter; the corresponding position of a post within the embedding space of a domain-adapted language model offers a second, independent source of evidence for or against a given pseudo-label. Confidence-based and neighbourhood-based filtering are therefore treated in this thesis as complementary rather than alternative criteria, and combined jointly for the self-training approach (Section 4.4).

3.3.2. Graph-Based Semi-Supervised Learning and Label Spreading

A second major family of SSL methods works from the opposite direction: instead of trusting the classifier’s own predictions, graph-based methods trust the geometric structure of the data. Both labelled and unlabelled observations are represented as nodes of a graph $G = (V, E)$, where an edge between two nodes encodes their similarity, typically computed from a k -nearest-neighbour relation or a kernel over the embedding space. Label information is then propagated from labelled to unlabelled nodes along the graph. This view is closely tied to the manifold assumption: if the data lie on or near a lower-dimensional structure embedded in a higher-dimensional space, labels are expected to vary smoothly along that structure, so that two observations that are close according to the intrinsic geometry of the data – rather than the raw feature space – should carry similar labels. Belkin, Niyogi, and Sindhwani (2006) formalise this intuition as manifold regularisation, casting the smoothness assumption as a reproducing-kernel-Hilbert-space regulariser that encompasses both transductive graph methods and their out-of-sample extensions, and providing the broader theoretical grounding for the graph-based algorithms that followed.

The specific algorithm used in this thesis, label spreading, follows the local and global consistency principle introduced by Dengyong Zhou et al. (2004): a good classification function should assign similar labels not only to nearby points (local consistency) but also to points lying on the same overall structure, such as a cluster or manifold (global consistency), going beyond what a plain nearest-neighbour rule would capture. In practice, the algorithm iteratively diffuses label information across the similarity graph while remaining anchored to the original labelled observations, with the density of the graph (how many neighbours each node is connected to) and the strength of the diffusion jointly controlling how far and how aggressively labels are allowed to spread. The output is not a hard label but a soft label distribution for every node, reflecting both the initial supervision and the degree of agreement among similar observations in the embedding space – a property that fits naturally with data, such as short opinionated text, where individual observations may be ambiguous but their position relative to other observations is informative. The

full formalisation of the propagation rule and of the role of its parameters is deferred to the Methodology chapter, where it is presented alongside the exact configuration used for Approach 4.

Applications to Sentiment Classification

Label propagation and label spreading have already been applied to sentiment classification outside the financial domain, with encouraging results. Ren et al. (2011) show that label propagation is more stable than SVM- and transductive-SVM-based alternatives for document-level sentiment classification in resource-scarce languages, where labelled data is especially scarce. Lu et al. (2022) propagate sentiment labels over a tripartite graph connecting reviews, product features, and opinion words in e-commerce reviews, exploiting relations beyond simple pairwise text similarity. Ledwaba and Marivate (2022) apply a graph-based semi-supervised approach to label political sentiment on Twitter data collected around local government elections, a setting that, like financial social media, combines short, informal text with a rapidly evolving topic. Together, these applications confirm that the smoothness assumption underlying label spreading transfers reasonably well to noisy, opinionated, and domain-specific text, which motivates its use as a second strategy, alongside self-training, for exploiting unlabelled financial social media posts in this thesis.

3.3.3. Synthesis: Two Complementary Strategies for Financial Sentiment

Self-training and label spreading operationalise the same underlying assumptions – smoothness and cluster structure – through different mechanisms for generating and filtering pseudo-labels, and each has already been validated, separately, on general-purpose sentiment classification. Self-training is model-driven: it trusts the classifier’s own predictions, refining that trust through confidence, embedding-space consistency, or both, as reviewed above. Label spreading is structure-driven: pseudo-labels are inferred from the geometry of a similarity graph built over labelled and unlabelled observations, rather than from the classifier’s own confidence. In this thesis, however, both mechanisms feed the same downstream step: the pseudo-labels each produces are used to enlarge the labelled set on which FinBERT itself is subsequently retrained, so the two strategies are better understood as two alternative ways of generating pseudo-labels for the same retraining procedure than as one approach that retrains a classifier and one that does not. What is missing from the literature reviewed above is a direct comparison between the two pseudo-labelling mecha-

nisms on the same dataset, under the same embedding space, and specifically for financial social media text, where short, informal, and rapidly shifting posts pose a harder test for both the confidence estimates used by self-training and the similarity graph used by label spreading than the review or forum text typically studied in prior work. This thesis addresses that gap by evaluating self-training with joint confidence-and-neighbourhood pseudo-label filtering (Approach 3) against label spreading (Approach 4) as two alternative routes to the same retrained FinBERT classifier, and by explicitly tuning both towards higher pseudo-label quality rather than higher pseudo-label quantity – a trade-off that, as the results in Chapter ?? show, is central to whether either strategy ultimately improves on a purely supervised baseline.

3.4. Market Sentiment and Financial Markets

Beyond the linguistic and modelling problem of extracting sentiment from text, a separate and older question is whether the resulting signal has any bearing on financial markets at all. The empirical literature that addresses this question distinguishes, at least implicitly, between three related but distinct constructs. *Sentiment* in the narrow sense refers to the directional tone of a text – bullish or bearish, optimistic or pessimistic. *Disagreement*, or dispersion, refers to the extent to which market participants hold different views about the same asset at the same time, regardless of the average tone. *Attention* refers to how much a given asset is being discussed at all, independently of what is being said about it. As this section shows, these three constructs have accumulated different, and at times contradictory, empirical support, and the distinction between them becomes especially important once financial social media data specifically – rather than news or analyst reports – is used as the textual source.

The foundational empirical demonstration that textual tone carries information for markets is due to Tetlock (2007), who quantifies the pessimism of the Wall Street Journal’s “Abreast of the Market” column using a general-purpose psychological dictionary and shows that unusually high media pessimism forecasts downward pressure on market prices, followed by a reversion towards fundamentals, and that both unusually high and unusually low pessimism predict higher trading volume. Bollen, Mao, and Zeng (2011) extend the same logic to social media, aggregating collective mood states expressed on Twitter and reporting that specific mood dimensions, most notably calmness, Granger-cause movements in the Dow Jones Industrial Average a few days later. This result has been widely cited but also widely contested on methodological grounds, including the stability of the mood-extraction dictionary and the robustness of the reported lag structure; it is discussed here less as a settled finding than as the paper that established

Twitter-derived sentiment as a legitimate object of financial inquiry.

3.4.1. Social Media, Disagreement, and Volatility

A more direct precedent for the present thesis is Antweiler and Frank (2004), who classify the content of Yahoo Finance and Raging Bull stock message boards into buy, sell, and hold categories using a Naive Bayes classifier and construct an aggregate bullishness index for each stock-day. Their central finding is that message-board activity helps predict market volatility and trading volume, while its effect on returns, although statistically detectable, is economically small and does not survive simple trading-strategy tests. This asymmetry between a strong volatility relationship and a weak or negligible return relationship is one of the most consistently replicated results in the finance-and-text literature, and it provides a direct empirical precedent for the forecasting comparison carried out later in this thesis, on a different data source, a different class of sentiment model, and two decades later. Cookson and Niessner (2020) revisit disagreement itself as the object of study, using self-reported bullish/bearish tags from StockTwits to measure investor disagreement directly rather than inferring it indirectly from message volume, and decompose that disagreement into disagreement about which information is relevant and disagreement in how the same information is interpreted. Their results indicate that periods of high disagreement coincide with higher trading volume and, again, greater volatility rather than a clear directional price effect – reinforcing the idea that dispersion in sentiment, not its average level, is what tends to move quantities associated with market activity and risk.

3.4.2. Attention as a Distinct Channel

A separate strand of work argues that even a purely tone-free measure of how much investors are paying attention to a stock carries predictive information. Da, Engelberg, and Gao (2011) use Google search volume for stock tickers as a direct, sentiment-free proxy for investor attention and show that spikes in search volume predict a short-term increase in the price of the corresponding stock, followed by a partial reversal over the following weeks – a pattern consistent with temporary price pressure from retail investors acting on newly acquired attention rather than on new fundamental information. This distinction matters for social-media-based studies specifically, because the volume of posts about a stock is simultaneously a measure of sentiment intensity, of disagreement (more posts typically means more distinct viewpoints), and of raw attention, and these three interpretations are easy to conflate if the sentiment level itself is treated as the only signal of interest.

Read together, this literature supports a more specific claim than the broad statement that “sentiment matters for markets.” The strongest and most consistently replicated relationships are not between sentiment level and future returns, but between measures of dispersion, volume, or attention on one side, and volatility or trading activity on the other; the direct link from sentiment to future returns is comparatively fragile and appears to depend heavily on the textual source, the market segment, and the time horizon considered. This pattern motivates treating volatility, rather than returns, as a primary outcome of interest when evaluating whether financial social media sentiment carries incremental forecasting value, and it motivates decomposing the sentiment signal used in this thesis into separate level, dispersion, and volume-based feature families rather than relying on a single aggregate sentiment score – a distinction taken up again in Section 5, where these feature families are evaluated separately against a purely financial baseline.

3.5. Deep Learning Models for Financial Forecasting

Before the arrival of deep learning, financial time series forecasting – volatility forecasting in particular – was dominated by autoregressive econometric models. Engle (1982) introduces the Autoregressive Conditional Heteroskedasticity (ARCH) model, which lets the variance of a time series depend on its own past squared shocks rather than assuming it is constant, and Bollerslev (1986) generalises it to GARCH by adding a dependence on past variance estimates as well. These models remain a standard benchmark against which any newer volatility forecasting method, including the machine-learning approach adopted in this thesis, is implicitly compared, precisely because they are simple, well understood, and difficult to beat out of sample. Deep learning entered this space primarily through recurrent architectures capable of modelling sequential dependence directly from data rather than through a fixed parametric recursion. Fischer and Krauss (2018) provide one of the most widely cited large-scale demonstrations of this shift, showing that Long Short-Term Memory (LSTM) networks trained to predict whether S&P 500 constituents will out- or underperform the cross-sectional median return outperform random forests, deep neural networks, and logistic regression on the same task and data. Sezer, Gudelek, and Ozbayoglu (2020) survey the broader wave of deep learning applications to financial time series between 2005 and 2019, covering price prediction, trend classification, and portfolio construction tasks, and note a recurring methodological weakness across much of this literature: evaluation protocols that do not fully guard against overfitting or information leakage between training and test periods, a concern directly addressed in this thesis through the walk-forward validation protocol described in the Methodology

chapter.

A second line of work asks whether textual signals, rather than price history alone, add forecasting value once combined with numerical market features. Xing, Cambria, and Welsch (2018) survey this literature broadly, covering inputs ranging from news articles and analyst reports to social media, and outcomes ranging from returns to volatility and risk measures, and conclude that reported performance gains are highly sensitive to the textual source, the target variable, and the evaluation horizon considered – a conclusion consistent with the mixed evidence already discussed in Section 3.4. Hu et al. (2018) offer a concrete architecture for this kind of fusion, using a hierarchical attention network to weight sequences of news representations before combining them with price information for stock trend prediction, and explicitly addressing the temporal alignment between when a piece of text becomes available and when the corresponding market outcome is realised. This alignment problem – ensuring that a feature could plausibly have been known to a trader at the time the prediction is made – recurs throughout the text-and-forecasting literature and is treated as a first-order methodological concern in this thesis as well, rather than as an afterthought.

3.5.1. Volatility Forecasting and the VIX

Forecasting volatility first requires deciding how to measure it, which is itself an active area of research rather than a settled convention. Garman and Klass (1980) show that estimators based on the full daily price range (open, high, low, close) extract substantially more information about the true underlying volatility than the simple squared close-to-close return typically used as a default, motivating the use of range- or return-based realised volatility measures with care rather than by default convention. Once volatility is measured, the GARCH family introduced above remains a strong econometric benchmark, but the financial market itself produces a competing, forward-looking estimate: the CBOE Volatility Index (VIX), derived from the prices of S&P 500 index options and widely regarded as the market’s own real-time estimate of expected near-term volatility. This makes the VIX an unusually demanding benchmark for any competing predictor, including a text-based one, since it already aggregates the information of a large and liquid options market rather than a single data source or model. Bai and Cai (2024) show that even the VIX’s own future values can be forecast more accurately using adaptive machine learning methods than with simpler benchmarks, indicating that volatility forecasting, including of the VIX itself, remains an open methodological problem rather than one where market-implied measures make further modelling redundant. For this reason, the VIX is treated in this thesis not only as a candidate outcome but, more importantly, as a control that

any claimed sentiment-based improvement in volatility forecasting must survive.

Taken together, the literature reviewed in this chapter suggests two gaps that this thesis is positioned to address. First, studies of text-based financial forecasting rarely test their claims against a financial baseline that already includes the market’s own best volatility estimate, which risks overstating the marginal value of a textual signal that a simple, already available market indicator could explain just as well. Second, this literature typically evaluates a single aggregate sentiment score rather than separating the level, dispersion, and volume-based constructs discussed in Section 3.4, even though the evidence reviewed there suggests these constructs relate very differently to returns, volatility, and trading activity. The evaluation protocol adopted in this thesis, detailed in Chapter 4, is designed specifically to close both gaps at once.

3.6. Chapter Summary

The first half of this chapter followed a single problem from raw text to a usable training signal. Representing financial social media text numerically is complicated by its informal register, addressed at the tokenisation and encoding level by subword schemes that keep abbreviated and non-standard tokens usable rather than discarding them (Section 3.1). Financial sentiment analysis further requires domain adaptation, whether through hand-built dictionaries, benchmarks such as the Financial PhraseBank, or fine-tuned transformer models such as FinBERT (Section 3.2). Because reliable sentiment labels for financial social media specifically remain scarce even by the standards of financial NLP, this thesis turns to semi-supervised learning, comparing a model-driven strategy – self-training with joint confidence-and-neighbourhood pseudo-label filtering – against a structure-driven one – label spreading over a similarity graph – two families that have each been validated separately on general-purpose sentiment classification, but never compared directly on financial social media text under the same embedding space (Section 3.3).

The second half of the chapter turned to whether the resulting sentiment signal has any bearing on markets. The empirical literature on this question distinguishes sentiment level, disagreement, and attention, and consistently finds the strongest and most replicated relationships not between sentiment and future returns, but between these other constructs and volatility or trading activity (Section 3.4). Deep learning models capable of exploiting this relationship, whether recurrent or transformer-based, have delivered clear gains over classical econometric benchmarks such as GARCH, but text-based forecasting studies specifically tend not to test their claims against the market’s own best volatility estimate, the VIX, nor to decompose the sentiment signal into its constituent constructs when doing so (Section 3.5).

Read together, these two halves point to a single underlying question, addressed at two different stages of the same pipeline. At the labelling stage: is a more carefully filtered, or a more structurally informed, pseudo-labelling strategy worth its added complexity relative to a simpler baseline, once labelled data for financial social media is genuinely scarce? At the forecasting stage: does a sentiment signal derived from that labelling process add value once evaluated against the toughest available financial baseline, and does the answer depend on which construct – level, dispersion, or volume – and which outcome – returns or volatility – is considered? This thesis is organised to answer both questions in turn, with the exact evaluation protocol, feature sets, and models used to do so detailed in Chapter 4.

4 | Model Design and Methodology

This chapter presents the full modelling pipeline developed for sentiment-based financial forecasting. The objective is to systematically evaluate different approaches for extracting sentiment signals from textual data and assess their potential predictive value.

We begin by describing the data preparation process, followed by four progressively more sophisticated modelling approaches. Each approach is motivated both from a theoretical and practical perspective.

4.1. Data Preparation

The dataset consists of 3,261,867 StockTwits posts referencing the SPY exchange-traded fund, which tracks the S&P 500 index, collected between 23 March 2020 and 6 April 2022 – a span of approximately two years. Each post is timestamped to the second and, at the time of posting, its author may optionally attach a self-reported sentiment tag: *Bullish* or *Bearish*. No *Neutral* tag exists in the raw data: a post either carries one of these two self-reported labels or none at all. Of the full dataset, 756,235 posts (23.2%) are labelled *Bullish*, 663,476 (20.3%) are labelled *Bearish*, and the remaining 1,842,156 (56.5%) carry no sentiment tag. Daily post volume varies considerably, averaging around 4,512 posts per day but ranging from as few as 30 to as many as 16,462 on the busiest days, reflecting bursts of attention around volatile market periods; this daily volume is itself later used as a feature in the forecasting stage of this thesis.

The primary goal of preprocessing is to transform this raw textual data into a structured representation suitable for machine learning models, while preserving relevant financial information. The preprocessing pipeline includes cleaning operations such as the removal of URLs, normalisation of whitespace, and standardisation of specific tokens. In particular, financial markers such as cashtags (e.g. \$AAPL), hashtags, and user mentions are preserved in a normalised form, as they carry important contextual information.

Only the 1,419,711 posts labelled *Bullish* or *Bearish* are used for supervised evaluation; the remaining unlabelled posts are set aside and exploited only by the semi-supervised approaches described later in this chapter (Approaches 3 and 4). A stratified train-test split, applied to the labelled subset only, allocates 80% (1,135,768 posts) to training and

20% (283,943 posts) to testing, preserving the class balance in both partitions. This fixed split is used for evaluation throughout the chapter, including for the zero-shot baseline in Approach 1.

4.1.1. Model Background: FinBERT

All approaches presented in this chapter build on FinBERT, a transformer-based language model adapted specifically for financial text. The general mechanics of the transformer architecture – self-attention, contextual token representations, and the pre-train-then-fine-tune paradigm – were introduced in Section 3.1.3 and are not repeated here; this section instead focuses on the specific model used and how it is applied in practice.

The checkpoint used throughout this thesis is `ProsusAI/finbert`, released by Araci (2019) and distributed via HuggingFace. Araci (2019) further pre-train a general-purpose BERT model (Devlin et al., 2019) on Reuters TRC2-financial, a domain-specific corpus of approximately 1.8 million financial news articles, before fine-tuning the resulting model for sentiment classification on the Financial PhraseBank (Malo et al., 2014), a set of financial news sentences manually annotated as *positive*, *neutral*, or *negative*. The domain rationale for this two-stage adaptation – why financial text requires more than a general-purpose sentiment model – is discussed in Section 3.2; as released, FinBERT is already specialised for financial sentiment and can be applied directly to new data or further adapted, as done in the approaches described below.

In practice, passing a text through FinBERT yields a probability distribution over the three sentiment classes. The [CLS] token embedding produced by the final transformer layer serves as an aggregate representation of the entire input sequence and is used as the input to the classification head throughout this chapter.

4.2. Approach 1: Zero-Shot FinBERT

As a starting point, the first approach applies FinBERT directly to the dataset, with none of its parameters updated and no task-specific adaptation. This provides the baseline against which all subsequent approaches are compared. The role of this baseline is methodological: if a more sophisticated approach cannot improve upon a strong pretrained model used out of the box, the additional complexity involved is difficult to justify.

The rationale for this baseline lies in transfer learning. Since FinBERT has already been exposed to large amounts of domain-relevant language during pretraining (Section 3.2), it encodes substantial knowledge of financial terminology, sentiment expressions, and market-related discourse, and may therefore produce meaningful predictions

even when applied to new, unseen data without fine-tuning. This makes zero-shot FinBERT a natural and computationally inexpensive starting point.

At the same time, this baseline is not expected to be perfectly aligned with the present task. The data used in this thesis consists of StockTwits posts, which differ substantially from the financial news text FinBERT was originally fine-tuned on: StockTwits text is short, informal, noisy, and rich in platform-specific conventions such as cashtags, abbreviations, slang, and speculative expressions. Even though FinBERT is domain-specific in a broad sense, a clear distribution shift therefore remains between its original fine-tuning data and the target data considered here.

A second, more specific issue concerns the label space. FinBERT is trained as a three-class sentiment classifier, producing *positive*, *neutral*, and *negative* predictions, whereas the present task is a binary problem with labels *Bullish* and *Bearish*. These concepts are related but not identical: a bullish post often expresses positive sentiment, but more specifically reflects an expectation of price appreciation, and analogously for bearish posts and price decline. To resolve this mismatch, the neutral class is discarded at decision time and the predicted label is instead taken as whichever of *positive* or *negative* receives the higher probability. This binary-forced decision rule keeps the model entirely zero-shot – no parameters are updated – while enforcing the same two-class decision structure used by all subsequent supervised and semi-supervised approaches, and avoids penalising the model for a three-way distinction the task never asked it to make.

Evaluation is performed only on the subset of posts with reliable ground-truth labels, namely those annotated as *Bullish* or *Bearish*; unlabelled posts are excluded from the classification metrics because they provide no trustworthy supervision, although all posts are still scored when constructing the aggregated daily sentiment signal used in the downstream forecasting pipeline.

4.2.1. Evaluation

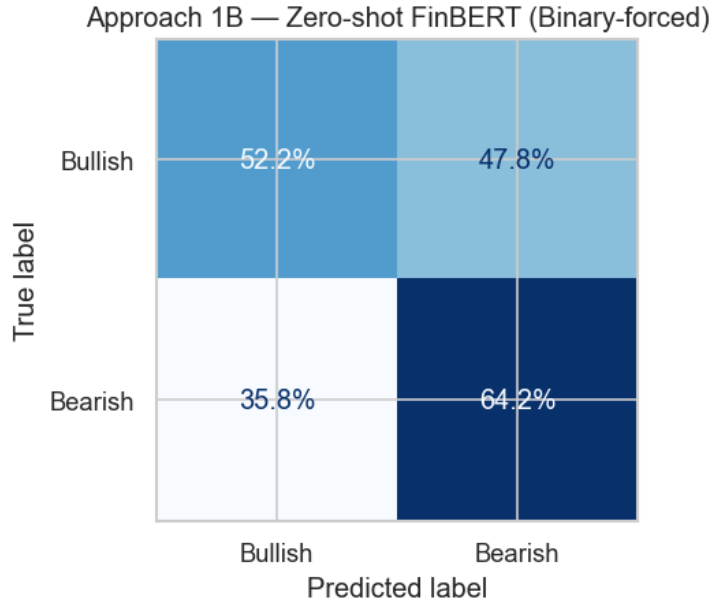


Figure 4.1: Confusion matrix for Approach 1 (zero-shot FinBERT, binary-forced decision rule).

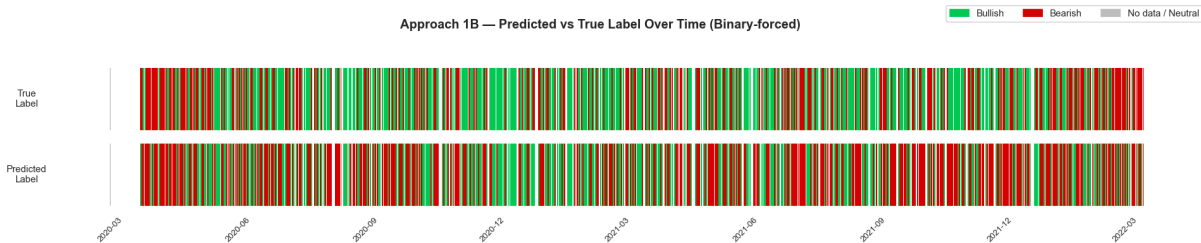


Figure 4.2: Daily sentiment score – true vs. predicted (Approach 1).

Class	Precision	Recall	F1-score
Bullish	0.54	0.64	0.59
Bearish	0.62	0.52	0.57

Table 4.1: Class-wise performance for zero-shot FinBERT (Approach 1).

4.2.2. Results and Interpretation

Zero-shot FinBERT achieves an accuracy of 0.5780 and a macro F1 score of 0.5778 – only marginally above what a random or majority-class baseline would achieve on a two-class

task. This confirms that FinBERT extracts a non-trivial signal even without adaptation, consistent with its pretraining on financial text, which allows it to capture general sentiment patterns relevant to market discourse.

Performance is nonetheless uneven across classes. The model recalls *Bullish* posts comparatively well (recall 0.64) but with lower precision (0.54), while for *Bearish* posts the pattern is reversed (precision 0.62, recall 0.52). This asymmetry indicates a mild bias towards predicting *Bullish* even where the true sentiment is *Bearish*, rather than a symmetric level of confusion between the two classes.

Taken together, these results confirm that zero-shot FinBERT provides a useful but imperfect baseline. The model captures some general sentiment polarity out of the box, but this polarity does not translate reliably into the specific directional signal required here, most likely because StockTwits posts are shorter, more informal, and more idiosyncratic than the financial news text on which FinBERT was fine-tuned. It offers a realistic lower bound on performance while exposing the limitations that motivate the dataset-specific fine-tuning introduced next (Approach 2) and the semi-supervised extensions that follow (Approaches 3 and 4).

4.3. Approach 2: Supervised Fine-Tuning

The second approach builds upon the zero-shot baseline by fine-tuning the pretrained FinBERT model on labelled examples drawn directly from the dataset. Rather than relying on FinBERT’s general financial knowledge alone, this approach explicitly adapts the model’s parameters to the specific task of classifying StockTwits posts as *Bullish* or *Bearish*, allowing it to adjust its representations to the language and labelling scheme of financial social media text.

This approach serves as the main supervised benchmark of the study. It is also particularly relevant because the semi-supervised approaches that follow rely on its quality: Approach 3 uses it to generate pseudo-labels, while Approach 4 relies on its learned representations for graph-based label spreading.

4.3.1. Training Data Selection

A key design decision concerns which observations to include in the training set. As established in Section 4.1, only 1,419,711 of the 3,261,867 posts in the dataset carry a self-reported *Bullish* or *Bearish* tag; the remainder carry no sentiment tag at all, since no *Neutral* label exists in the raw data. Assigning an artificial label to these untagged posts would introduce noise and distort the model’s decision boundary. For this reason,

only observations labelled *Bullish* or *Bearish* are retained for fine-tuning, and the task is framed as a binary classification problem between the two.

4.3.2. FinBERT Architecture

FinBERT is built on the BERT-base architecture, comprising an embedding layer, a stack of Transformer encoder blocks, a pooler, and a final classification head, as illustrated in Figure 4.3. An input post is first tokenised and embedded, then passed through the stack of encoder blocks; the internal structure of a single block, repeated identically twelve times, is shown in Figure 4.4. The specific configuration used throughout this thesis is:

- Hidden size: 768
- Number of Transformer encoder layers: 12
- Number of attention heads: 12
- Maximum sequence length used: 128 tokens

As described in Section 4.1.1, the [CLS] token embedding from the final layer is used as the input to the classification head; for this approach, the classification head is re-initialised and trained from scratch on the Bullish/Bearish task.

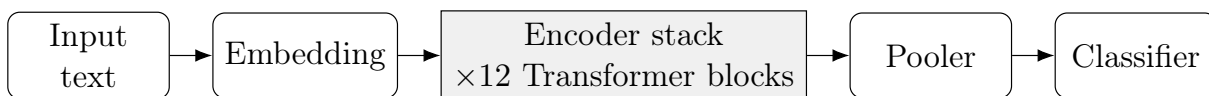


Figure 4.3: High-level FinBERT pipeline, from raw input text to the final Bullish/Bearish prediction.

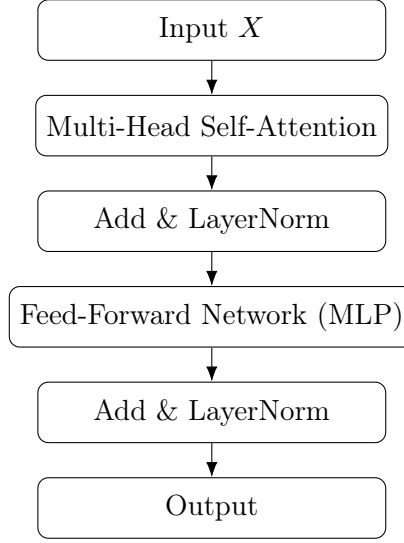


Figure 4.4: Internal structure of a single Transformer encoder block, repeated 12 times in FinBERT (cf. Section 3.1.3).

4.3.3. Fine-Tuning Strategy: Progressive Unfreezing

The fine-tuning strategy adopted is progressive unfreezing, in which the model is trained sequentially in three phases, starting from the most task-specific parameters and progressively allowing deeper encoder layers to update. This gradual schedule is preferred over unfreezing the entire model from the start because large, immediate updates to already highly informative pretrained representations risk catastrophic forgetting; adapting the classification head first, and only then progressively exposing deeper layers to smaller learning rates, is intended to preserve FinBERT’s pretrained financial-language knowledge while still allowing it to specialise to the target task.

In the first phase, all FinBERT encoder layers are frozen and only the classification head is trained, so that the classifier learns the Bullish/Bearish decision boundary from fixed FinBERT representations. Subsequently, the pooler and the last two Transformer encoder layers are unfrozen together with the classifier, allowing the upper layers to adapt to the task while the lower layers, which encode more general linguistic information, are preserved. In the final phase, all layers are unfrozen and the full model is fine-tuned end to end.

The learning-rate schedule used across the three phases is:

$$5 \times 10^{-4} \rightarrow 2 \times 10^{-5} \rightarrow 1 \times 10^{-5}$$

and the epoch budget is divided across the three phases as:

$$20\% / 30\% / 50\%$$

Training is monitored on the validation set at every phase, and early stopping is applied to retain the checkpoint with the best validation macro F1-score.

4.3.4. Evaluation

The selected model is evaluated on the held-out Bullish/Bearish test set described in Section 4.1. The evaluation includes the confusion matrix, predicted versus true labels over time, and class-wise classification metrics.

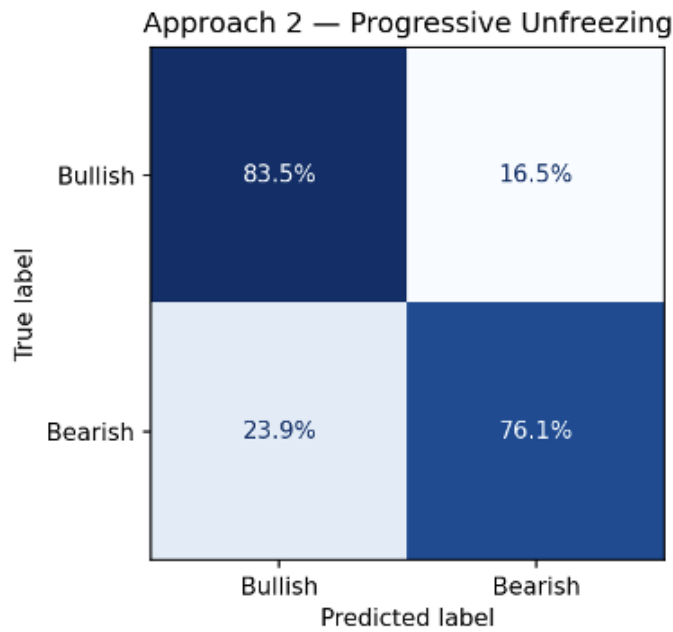


Figure 4.5: Confusion matrix – Approach 2 (Progressive Unfreezing).

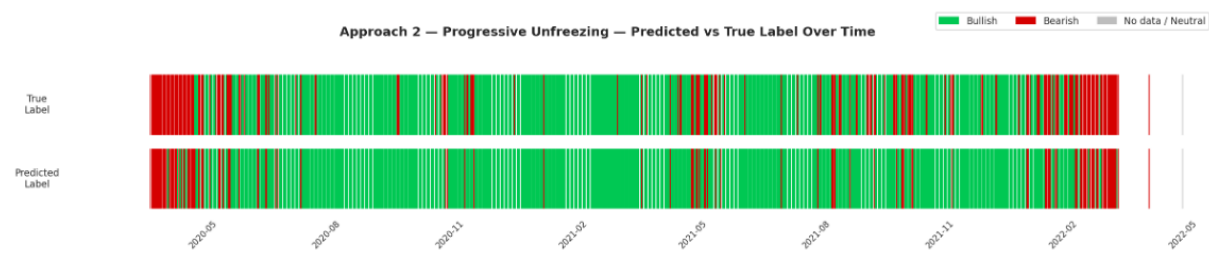


Figure 4.6: Predicted vs. true labels over time – Approach 2.

Class	Precision	Recall	F1-score	Support
Bullish	0.80	0.84	0.82	151247
Bearish	0.80	0.76	0.78	132696
Macro Avg	0.80	0.80	0.80	283943
Weighted Avg	0.80	0.80	0.80	283943

Table 4.2: Performance metrics – Approach 2 (Progressive Unfreezing).

4.3.5. Discussion

Fine-tuning improves over the zero-shot FinBERT baseline because the model is no longer used as a generic financial sentiment classifier: it is instead adapted to the specific Bullish/Bearish labels and to the language distribution of the dataset. This adaptation is especially relevant for financial social media text, where posts are short, informal, and often contain market-specific expressions; by updating the model’s parameters on labelled examples from the target dataset, fine-tuning allows FinBERT to learn a more appropriate decision boundary for this specific task.

This gradual approach reflects a deliberate trade-off in favour of training stability: unfreezing the entire model from the start would make early training more sensitive to the learning-rate choice and more likely to overwrite useful pretrained representations before the classification head has learned a reasonable decision boundary. Monitoring validation macro F1-score at each phase and applying early stopping ensures that the retained checkpoint is selected according to generalisation performance rather than training loss alone.

Overall, this supervised fine-tuning step provides a stronger baseline than zero-shot inference and forms the foundation for the semi-supervised approaches that follow.

4.3.6. Strengths and Limitations

The motivation for this approach is grounded in the limitations of the zero-shot baseline discussed in Section 4.2: FinBERT’s pretraining distribution is unlikely to align fully with StockTwits text, which is short, noisy, and rich in platform-specific conventions such as cashtags, abbreviations, and informal slang, and its native output space (*positive*, *neutral*, *negative*) does not align precisely with the task’s label space (*Bullish*, *Bearish*).

Fine-tuning addresses both issues simultaneously. By updating the model’s parameters on in-domain labelled examples, the model learns to map the specific linguistic patterns of StockTwits to the correct Bullish/Bearish labels; re-initialising and training the classification head from scratch for the binary task eliminates the semantic mismatch

present in the zero-shot setting, while updating the deeper Transformer layers lets the model refine its internal representations to the actual data distribution.

Despite these advantages, supervised fine-tuning has one important structural limitation: it makes no use of the unlabelled portion of the dataset. The majority of StockTwits posts carry no sentiment tag and are therefore excluded from training entirely, even though they carry potentially valuable information about the linguistic structure of the data, the distribution of topics, and the semantic space in which Bullish and Bearish posts live. By discarding this unlabelled pool entirely, Approach 2 leaves a potentially significant source of signal unexploited – a limitation that motivates the semi-supervised extensions developed in Approaches 3 and 4, which attempt to incorporate unlabelled data into the learning process in a principled way.

4.4. Approach 3: Semi-Supervised Self-Training

The third approach explores a semi-supervised self-training strategy to leverage the large amount of unlabelled data available in the dataset. The core idea is to use the model trained on labelled data in Approach 2 to generate pseudo-labels for unlabelled observations, filter these pseudo-labels for reliability, and retrain the model on the resulting, larger dataset. The motivation is straightforward: the majority of posts in the dataset carry no sentiment tag at all (Section 4.1), and Approach 3 attempts to extract useful supervisory signal from this pool instead of discarding it entirely, under the hypothesis that it encodes structural information about the data distribution that can improve generalisation. Unlike purely supervised learning, this approach allows the model to exploit additional information contained in unlabelled data, while maintaining control over noise through confidence-based and structure-based filtering mechanisms.

4.4.1. Theoretical Justification and Assumptions

The effectiveness of Approach 3 rests on two core assumptions. First, that the fine-tuned Approach 2 model is accurate enough that its high-confidence predictions on unlabelled data are predominantly correct. Second, that the cluster assumption holds in the embedding space produced by the fine-tuned model – that is, that semantically similar posts tend to share the same sentiment label. Both assumptions are plausible given that the Approach 2 embeddings are already adapted to the task and dataset, but neither is guaranteed; if the labelled training set used to produce the Approach 2 model is itself unrepresentative, the pseudo-labels derived from it may not be useful. Section 4.4.3 validates this assumption by assessing the pipeline’s ability to recover labels whose ground

truth is known but withheld during prediction.

4.4.2. Methodology

Step 1: Initial Pseudo-Labeling

The first step consists of running the fine-tuned FinBERT model from Approach 2 on the entire dataset, including both the labelled and unlabelled observations. For each post, the model produces a probability distribution over the two classes (*Bullish* and *Bearish*), from which the predicted label and the associated confidence score, defined as the maximum class probability, are extracted. Rather than being treated as ground truth, pseudo-labels are regarded as tentative predictions. Their confidence scores vary substantially across the unlabelled pool, motivating the filtering strategy described in the following two steps.

Step 2: Confidence-Based Filtering

To reduce noise, only high-confidence pseudo-labelled observations are retained. A threshold of $\text{CONF_TH} = 0.90$ is applied to the 1,842,156 originally unlabelled posts, retaining 932,384 pseudo-labelled observations (493,960 *Bullish*, 438,424 *Bearish*). This step ensures that only the most confident pseudo-labels proceed to the next, structure-based filtering stage.

Step 3: Structural Filtering via kNN Agreement

To further improve pseudo-label quality, a structure-based filtering step is introduced. Instead of relying solely on model confidence, this step evaluates the consistency of each pseudo-label with its local neighbourhood in embedding space. Embeddings are extracted from the *base* FinBERT model, prior to fine-tuning, so that the neighbourhood structure used for filtering is independent of the classifier that produced the pseudo-labels being checked. A k -nearest-neighbour graph is built over all samples, and for each pseudo-labelled point the agreement between its label and those of its neighbours is computed; only points with agreement of at least 0.80 are retained. This reduces the 932,384 confidence-filtered pseudo-labels to 383,738 – a retention rate of 41.2% relative to that pool.

This criterion operationalises the *cluster assumption*, a foundational principle in semi-supervised learning (Chapelle, Schölkopf, and Zien, 2006): points that are close to each other in the representation space are likely to belong to the same class. If a pseudo-labelled point is surrounded by neighbours carrying the same label, that label is more likely to

reflect genuine local structure in the data than an isolated model error; conversely, if a point’s predicted label disagrees with the majority of its neighbours, it may lie near a decision boundary or in a region of high uncertainty, and is discarded. The combination of confidence filtering and neighbourhood-agreement filtering makes Approach 3 a *selective* semi-supervised method: rather than attempting to exploit the entire unlabelled pool, it identifies a smaller, high-quality subset of unlabelled observations for which the pseudo-labels are likely to be both individually confident and structurally consistent.

Step 4: Construction of the Semi-Supervised Training Set

The final training set combines the original labelled data with the filtered pseudo-labelled samples. The labelled training split from Section 4.1 is itself divided into a true training set of 908,611 posts and a true validation set of 227,154 posts, so that the validation set used for early stopping contains only genuine labels and is never contaminated by pseudo-labels. Adding the 383,738 refined pseudo-labelled posts to the true training set yields a combined training set of 1,292,352 samples – roughly 42% larger than the purely supervised training set used in Approach 2.

Step 5: Retraining the Final Classifier

A new FinBERT model, using the same architecture as Approach 2 (Section 4.3), is trained from scratch on the combined dataset, with early stopping applied on the true-label validation set to prevent overfitting. This step allows the model to refine its decision boundary using evidence from both the original labelled examples and the filtered pseudo-labelled ones, rather than from the original labelled data alone.

4.4.3. Pseudo-Label Quality Validation

The filtering strategy in Steps 2–3 rests on the assumption that combining confidence and neighbourhood agreement selects pseudo-labels that are more often correct. This assumption is tested directly, rather than left implicit: 200,000 genuinely labelled posts are deliberately withheld from the pipeline and treated as if unlabelled, pseudo-labels are assigned to them by the same confidence-and-kNN procedure under a grid of threshold combinations, and the assigned pseudo-labels are compared against the true, withheld labels. Table 4.3 reports, for each combination of confidence threshold (CONF_TH) and minimum neighbourhood agreement (KNN_MIN), the number and share of the 200,000 samples retained, together with the precision, recall, and F1-score of the retained pseudo-labels against ground truth.

CONF_TH	KNN_MIN	N kept	% kept	Precision	Recall	F1
0.80	–	183,397	91.7%	0.8205	0.8203	0.8204
0.80	0.70	84,820	42.4%	0.8764	0.8707	0.8731
0.80	0.80	58,393	29.2%	0.8970	0.8892	0.8926
0.80	0.90	35,275	17.6%	0.9211	0.9126	0.9164
0.90	–	177,570	88.8%	0.8279	0.8278	0.8278
0.90	0.70	82,267	41.1%	0.8821	0.8771	0.8793
0.90	0.80	56,681	28.3%	0.9018	0.8951	0.8981
0.90	0.90	34,458	17.2%	0.9243	0.9168	0.9202
0.95	–	171,986	86.0%	0.8343	0.8342	0.8343
0.95	0.70	79,629	39.8%	0.8873	0.8828	0.8848
0.95	0.80	54,814	27.4%	0.9066	0.9007	0.9033
0.95	0.90	33,276	16.6%	0.9285	0.9224	0.9252
0.97	–	168,083	84.0%	0.8384	0.8383	0.8384
0.97	0.70	77,928	39.0%	0.8898	0.8858	0.8876
0.97	0.80	53,620	26.8%	0.9089	0.9036	0.9060
0.97	0.90	32,481	16.2%	0.9305	0.9250	0.9276
0.99	–	159,089	79.5%	0.8487	0.8487	0.8487
0.99	0.70	73,995	37.0%	0.8962	0.8927	0.8943
0.99	0.80	50,867	25.4%	0.9138	0.9092	0.9113
0.99	0.90	30,641	15.3%	0.9341	0.9296	0.9317

Table 4.3: Pseudo-label quality validation on 200,000 withheld labelled posts, across a grid of confidence thresholds and minimum kNN agreement levels. A dash in the KNN_MIN column indicates confidence-only filtering, with no neighbourhood-agreement step applied.

Two patterns emerge clearly. First, for any fixed confidence threshold, adding the kNN agreement filter and raising its minimum agreement level monotonically improves precision, recall, and F1 on the withheld labels, at the cost of retaining a much smaller fraction of the pool – confirming that confidence and neighbourhood agreement are complementary, not redundant, reliability signals. Second, the effect of the confidence threshold alone is comparatively small once the kNN filter is fixed: moving from CONF_TH = 0.80 to 0.99 at KNN_MIN = 0.90 improves F1 only from 0.9164 to 0.9317, whereas adding the kNN filter at a fixed confidence threshold produces a much larger jump (e.g. at CONF_TH = 0.90, F1 rises from 0.8278 with no kNN filter to 0.9202 at KNN_MIN = 0.90). The best configuration overall is CONF_TH = 0.99 with KNN_MIN = 0.90 (F1 = 0.9317), and this combination is used in the robustness check reported in Section 4.4.5.

4.4.4. Ablation Study: Isolating the Contribution of Each Filtering Step

The validation presented in the previous section assesses the reliability of the generated pseudo-labels. However, high pseudo-label accuracy does not necessarily translate into *better downstream classifier*: applying stricter filtering typically improves label quality, but also reduces the number of training samples. To evaluate this trade-off efficiently, a lightweight probe model, consisting of the last two FinBERT encoder layers, the pooler, and the classification head (approximately 7M of FinBERT’s 110M parameters), is re-trained on the same 908k/283k train/test split under different confidence threshold and kNN settings, avoiding the need to repeat the full retraining procedure of Step 5 for every configuration.

Configuration	N pseudo	% discarded	Val F1	Test Acc	Test F1
Baseline (no pseudo-labels)	0	100.0%	0.7373	0.7388	0.7375
CONF 0.70, no kNN	1,403,653	23.8%	0.7549	0.7568	0.7577
CONF 0.80, no kNN	1,199,099	34.9%	0.7567	0.7579	0.7577
CONF 0.70, KNN 0.70	700,869	62.0%	0.7433	0.7437	0.7436
CONF 0.80, KNN 0.70	623,127	66.2%	0.7461	0.7477	0.7472
CONF 0.70, KNN 0.80	506,555	72.5%	0.7438	0.7457	0.7449
CONF 0.80, KNN 0.80	458,239	75.1%	0.7424	0.7441	0.7432
CONF 0.70, KNN 0.90	329,591	82.1%	0.7383	0.7417	0.7399
CONF 0.80, KNN 0.90	303,957	83.5%	0.7436	0.7441	0.7433

Table 4.4: Ablation study on the lightweight FinBERT probe: contribution of confidence and kNN filtering to downstream test performance. The CONF 0.70, KNN 0.70 row is inferred from the source log (its original label was ambiguous between `conf70_knn70` and `conf80_knn70`, which appeared twice with different values; this labelling should be re-checked against the original results file before final submission).

This ablation shows that adding pseudo-labels always helps relative to the purely supervised baseline (macro F1 ≈ 0.737 with no pseudo-labels at all), confirming that the unlabelled pool does carry useful signal. However, two further findings complicate the picture. First, within this lightweight-probe setting, the configuration with the *lowest* confidence threshold and no kNN filter – which retains the largest number of pseudo-labels – achieves the highest test macro F1 (≈ 0.759), suggesting that at this scale, pseudo-label *quantity* contributes more to downstream performance than additional strictness. Second, adding the kNN agreement filter on top of confidence filtering slightly *reduces* test F1 relative to confidence filtering alone, and a very strict kNN threshold performs little better than the no-pseudo-label baseline. Taken together with Section 4.4.3, this indi-

cates that confidence and kNN filtering improve pseudo-label *correctness* monotonically, but do not straightforwardly improve downstream classifier performance in the same direction, because each additional filtering criterion also shrinks the training set. This trade-off between label quality and training-set size is examined directly, at full scale, in the robustness check that follows.

4.4.5. Robustness Check: Full-Scale Retraining at Quality-Optimised Thresholds

Motivated by the trade-off identified above, the full-scale pipeline (Steps 1–5, using the complete FinBERT model rather than the lightweight probe) is retrained twice more, using the confidence thresholds identified as most reliable in Section 4.4.3, both paired with $\text{KNN_MIN} = 0.90$.

Configuration	N pseudo	% of pool	Acc	F1	Bull. P/R	Bear. P/R
Base (0.90,0.80)	383,738	20.8%	0.8073	0.8067	0.79/0.81	0.83/0.81
CONF 0.97, KNN 0.90	576,861	31.3%	0.8015	0.8004	0.79/0.78	0.81/0.82
CONF 0.99, KNN 0.90	317,878	17.3%	0.8004	0.8000	0.77/0.81	0.83/0.79

Table 4.5: Robustness check: full-scale retraining at the confidence thresholds identified as most reliable in the pseudo-label quality validation, compared against the base configuration used as the canonical Approach 3 model. All three rows are evaluated on the same held-out test set ($n = 283,943$; Bullish support 132,696, Bearish support 151,247).

Both quality-optimised configurations perform marginally *worse* on the held-out test set than the base configuration reported in Section 4.4.6 ($\text{CONF_TH} = 0.90$, $\text{KNN_MIN} = 0.80$, 383,738 pseudo-labels, Accuracy = 0.8073, F1 macro = 0.8067), despite the base configuration using a laxer filter. This mirrors the pattern already observed in the ablation study: optimising purely for pseudo-label correctness does not guarantee a better final classifier, because stricter filtering reduces the size of the augmented training set, and at full scale this reduction outweighs the modest gain in label correctness. For this reason, the base configuration is retained as the canonical Approach 3 model used in the cross-approach comparison later in this chapter, and the two quality-optimised runs are reported here as a robustness check rather than as alternative candidates for that role.

4.4.6. Evaluation

The final model (base configuration: $\text{CONF_TH} = 0.90$, $\text{KNN_MIN} = 0.80$) is evaluated on the held-out test set described in Section 4.1, containing only genuinely labelled

observations.

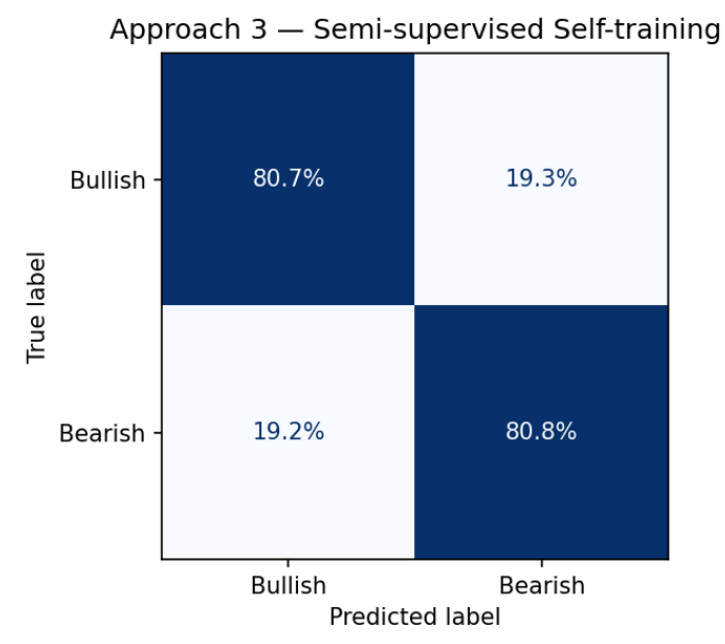


Figure 4.7: Confusion matrix – Approach 3.

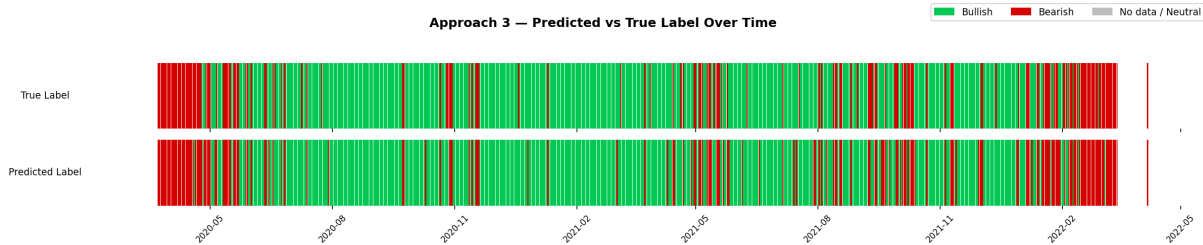


Figure 4.8: Daily sentiment score – true vs. predicted (Approach 3).

Class	Precision	Recall	F1-score	Support
Bullish	0.79	0.81	0.80	132696
Bearish	0.83	0.81	0.82	151247
Macro Avg	0.81	0.81	0.81	283943
Weighted Avg	0.81	0.81	0.81	283943

Table 4.6: Performance metrics – Approach 3 (Accuracy = 0.8073, F1 macro = 0.8067).

4.4.7. Discussion

Self-training leverages unlabelled data effectively when combined with appropriate filtering: confidence-based filtering ensures that only reliable pseudo-labels are used, while

the kNN agreement step adds a further layer of robustness by enforcing consistency with local data structure, together reducing the risk of confirmation bias, a well-known failure mode of self-training. The quality validation in Section 4.4.3 confirms that this combined filter does select more accurate pseudo-labels than confidence alone; at the same time, the ablation and robustness checks in the two sections that follow it show that this gain in label accuracy does not automatically translate into a better final classifier, since stricter filtering also reduces the amount of data available for retraining. The base configuration used throughout this thesis reflects this trade-off rather than the single most accurate pseudo-labelling configuration in isolation.

4.4.8. Strengths and Limitations

The self-training paradigm is one of the most widely used approaches in semi-supervised learning: a model trained on labelled data assigns pseudo-labels to unlabelled observations, which are then incorporated into the training set for further refinement, under the assumption that confident predictions are likely to be correct. A key strength of the implementation in this thesis is that this assumption is not left untested: the quality validation in Section 4.4.3 checks it directly against withheld ground truth, rather than relying on it purely as a theoretical justification.

Self-training nonetheless remains inherently exposed to *error propagation*: since pseudo-labels are treated as ground truth during retraining, incorrect predictions may be reinforced rather than corrected, particularly when the seed model is imperfect or when the unlabelled data distribution differs from the labelled one. The conservative selection strategy adopted here – requiring both high confidence and strong neighbourhood agreement – mitigates this risk but introduces its own trade-off: only a comparatively small share of the unlabelled pool is retained, and, as the ablation and robustness checks show, pushing this strictness further does not necessarily improve the final classifier. The method therefore sacrifices coverage for reliability, and the results in this section suggest that this trade-off has a practical limit beyond which further strictness stops paying for itself.

4.5. Approach 4: Graph-Based Label Spreading

The fourth approach adopts a different paradigm for exploiting unlabelled data: a graph-based semi-supervised method based on label spreading. Rather than relying on a classifier’s own confidence and local neighbourhood agreement to select individual pseudo-labels, as in Approach 3, this method constructs a similarity graph over all observations and diffuses known labels across it, allowing unlabelled observations to inherit label in-

formation from semantically similar instances anchored by the labelled nodes. As with Approach 3, the pseudo-labels produced by this process are ultimately used to retrain FinBERT (Section 3.3.3); the distinction between the two approaches lies in the pseudo-label generation and filtering process, while both retrain a classifier using the selected pseudo-labels.

4.5.1. Theoretical Foundation: The Manifold Assumption

The approach is grounded in the *manifold assumption*, one of the two core principles of semi-supervised learning (Chapelle, Schölkopf, and Zien, 2006). This assumption states that data, although observed in a high-dimensional space, lies approximately on a lower-dimensional manifold, and that points close to each other on this manifold are likely to share the same label. In the context of sentiment classification, this translates to the intuition that posts expressing similar ideas in similar language should carry the same sentiment orientation, regardless of whether they are labelled or not.

Label propagation algorithms construct a graph that approximates this manifold structure and then propagate known labels along the graph’s edges. The key difference from Approach 3 is that label spreading exploits the *global* geometry of the data manifold, rather than relying on local confidence estimates or neighbourhood-agreement checks applied independently to each point: where Approach 3 makes selective decisions at the level of individual observations, Approach 4 treats pseudo-label generation as a global optimisation over the entire data graph.

4.5.2. Methodology

Step 1: Embedding Extraction

Each post is mapped into a dense vector representation using the encoder of the fine-tuned model from Approach 2: the [CLS] token embedding is used as a compact, 768-dimensional representation of the entire text. The fine-tuned encoder is used, rather than base FinBERT, because it already captures domain-specific semantic and sentiment information, so that semantically and sentimentally similar posts are expected to lie close together in the resulting representation space.

Step 2: Normalisation and Dimensionality Reduction

The extracted embeddings are first L2-normalised, so that similarity comparisons depend only on direction rather than magnitude. Principal Component Analysis (PCA) is then

applied, projecting the 768-dimensional embeddings onto their first 50 principal components, which together explain 70.57% of the total variance. This step reduces noise and redundancy in the representation space, and improves the stability of the similarity graph constructed in the next step, since distance measures in very high-dimensional spaces become less reliable (the curse of dimensionality).

Step 3: Graph Construction

A similarity graph is constructed over all observations using a k -nearest-neighbour (kNN) approach: each node is connected to its k closest neighbours in the reduced embedding space, so that label propagation occurs primarily along locally similar points. The number of neighbours k controls the extent of information flow across the graph, with smaller values emphasising local structure and larger values allowing more global smoothing; the choice of k is tuned jointly with the spreading strength α in Section 4.5.3.

Step 4: Label Initialisation

The graph is initialised by assigning the known *Bullish*/*Bearish* labels to the labelled training observations, while all other nodes start with no class information (following the `scikit-learn` convention, this is implemented as zero probability mass across both classes rather than a uniform prior). These unlabelled nodes then receive a label distribution through the propagation process described next.

Step 5: Label Spreading

Label spreading propagates label information across the graph: each node iteratively updates its label distribution based on its neighbours, while remaining anchored to the originally labelled observations. The parameter α controls the strength of this diffusion at each iteration: higher values allow labels to propagate further and produce smoother, more globally consistent distributions, while lower values keep predictions closer to the original labels. The output of this step is a soft label distribution – a pair of class probabilities – for every node in the graph, including the originally labelled ones.

Step 6: Confidence-Based Pseudo-Label Selection

The soft label distributions produced by label spreading are converted into pseudo-labels only for originally unlabelled nodes whose highest class probability exceeds a confidence threshold `CONF_TH`; nodes below this threshold are left without a pseudo-label rather than being forced into either class, since the graph itself expresses genuine uncertainty

about them. This mirrors the role played by confidence-based filtering in Approach 3, but the confidence score here comes from the graph’s propagated label distribution rather than from a classifier’s own softmax output.

Step 7: Retraining the Final Classifier

As in Approach 3 (Section 4.4), the selected pseudo-labels are combined with the original labelled training data, and a new FinBERT model, using the same architecture as Approach 2, is trained from scratch on the combined dataset. This is the model ultimately evaluated in Section 4.5.6: label spreading in this thesis is used as a pseudo-labelling mechanism that feeds a retrained classifier, not as a stand-alone transductive classifier evaluated directly on its own soft predictions.

4.5.3. Hyperparameter Optimisation

A grid search over (k, α) is performed on an inner validation split derived from the labelled training set, with $k \in \{5, 10, 15, 20, 30\}$ and $\alpha \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$. The best configuration is $k = 10, \alpha = 0.7$: a moderately sized neighbourhood combined with strong diffusion, indicating that allowing labels to propagate relatively far through the graph is beneficial for this task. With (k, α) fixed at this configuration, a coarse scan of the confidence threshold over $\text{CONF_TH} \in \{0.50, 0.55, \dots, 0.80\}$ identifies $\text{CONF_TH} = 0.60$ as the best-performing value on the inner validation split, retaining 1,604,518 pseudo-labels (87.1% of the 1,842,156 unlabelled posts; 791,147 *Bullish*, 813,371 *Bearish*).

4.5.4. Pseudo-Label Quality Validation

As in Approach 3, the assumption that higher-confidence pseudo-labels are more often correct is tested directly: 200,000 genuinely labelled posts are withheld and treated as unlabelled, pseudo-labels are assigned to them using the tuned graph ($k = 10, \alpha = 0.7$) at a range of confidence thresholds, and the assigned pseudo-labels are compared against the true, withheld labels.

CONF_TH	N kept	% kept	Precision	Recall	F1
0.60	185,501	92.8%	0.8200	0.8205	0.8203
0.65	178,501	89.3%	0.8300	0.8306	0.8303
0.70	168,923	84.5%	0.8447	0.8452	0.8449
0.75	160,793	80.4%	0.8566	0.8569	0.8568
0.80	151,375	75.7%	0.8700	0.8707	0.8703
0.85	140,546	70.3%	0.8861	0.8865	0.8863
0.90	128,671	64.3%	0.8992	0.8998	0.8995
0.95	112,954	56.5%	0.9156	0.9163	0.9159
0.97	103,351	51.7%	0.9253	0.9260	0.9256
0.99	84,073	42.0%	0.9395	0.9404	0.9399

Table 4.7: Pseudo-label quality validation on 200,000 withheld labelled posts, across a range of confidence thresholds ($k = 10$, $\alpha = 0.7$ fixed).

As with Approach 3, pseudo-label correctness improves monotonically as the confidence threshold is raised, from $F1 = 0.8203$ at $CONF_TH = 0.60$ to $F1 = 0.9399$ at $CONF_TH = 0.99$, at the cost of retaining a progressively smaller share of the unlabelled pool. This motivates testing, in the robustness check that follows, whether retraining at a stricter, quality-optimised threshold produces a better final classifier than the threshold selected by the coarser grid search above.

4.5.5. Robustness Check: Full-Scale Retraining at Quality-Optimised Thresholds

The full-scale pipeline (Steps 1–7, with $k = 10$, $\alpha = 0.7$ fixed) is retrained twice more, at the higher confidence thresholds identified as most reliable in Section 4.5.4:

- $CONF_TH = 0.97$: 645,730 pseudo-labels retained (35.1% of the unlabelled pool)
– Accuracy = 0.8088, F1 macro = 0.8081.
- $CONF_TH = 0.99$: 445,715 pseudo-labels retained (24.2% of the unlabelled pool)
– Accuracy = 0.8076, F1 macro = 0.8070.

Unlike in Approach 3, where the base configuration outperformed both quality-optimised reruns, here $CONF_TH = 0.97$ outperforms the base configuration from Section 4.5.3 ($CONF_TH = 0.60$, 1,604,518 pseudo-labels, Accuracy = 0.8067, F1 macro = 0.8060), while the strictest configuration ($CONF_TH = 0.99$) again performs slightly worse than 0.97. This suggests that, for label spreading specifically, discarding the least-confident portion of the propagated pseudo-labels does improve the final classifier, but only up to a

point beyond which the loss of training volume outweighs the further gain in label correctness – the same quality-versus-quantity tension identified for Approach 3, but resolved in the opposite direction here. For this reason, $\text{CONF_TH} = 0.97$ is retained as the canonical Approach 4 model used in the cross-approach comparison later in this chapter.

4.5.6. Evaluation

The canonical model ($k = 10$, $\alpha = 0.7$, $\text{CONF_TH} = 0.97$) is evaluated on the held-out test set described in Section 4.1.

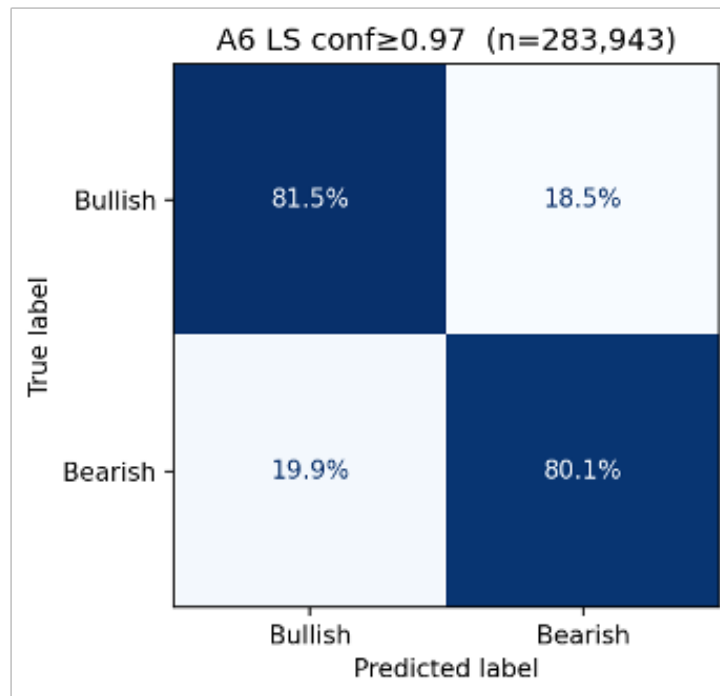


Figure 4.9: Confusion matrix – Approach 4 ($\text{CONF_TH} = 0.97$).

Class	Precision	Recall	F1-score	Support
Bullish	0.79	0.80	0.80	132696
Bearish	0.82	0.82	0.82	151247
Macro Avg	0.81	0.81	0.81	283943
Weighted Avg	0.81	0.81	0.81	283943

Table 4.8: Performance metrics – Approach 4 (Accuracy = 0.8088, F1 macro = 0.8081).

4.5.7. Discussion

Label spreading offers a complementary route to the same goal pursued by Approach 3: both ultimately retrain FinBERT on labelled data augmented with pseudo-labels, but they generate and filter those pseudo-labels differently – Approach 3 through a classifier’s own confidence and local neighbourhood agreement, Approach 4 through global diffusion over a similarity graph. The pseudo-label quality validation confirms that, as for Approach 3, stricter filtering produces more accurate pseudo-labels; the robustness check shows that for label spreading this translates into a genuinely better final classifier at $\text{CONF_TH} = 0.97$, unlike for Approach 3, where the laxer base configuration remained the strongest. Approach 4 also achieves the best result of all four approaches developed in this chapter, narrowly ahead of Approach 3, which is consistent with it exploiting the global structure of the embedding space rather than making purely local, point-by-point decisions.

4.5.8. Strengths and Limitations

Approach 4 has some advantages relative to the preceding methods. It is the only approach that considers the geometry of the *entire* unlabelled pool jointly, through the similarity graph, rather than filtering candidate pseudo-labels one at a time; and because pseudo-label generation is a closed-form graph computation, it decouples the (comparatively cheap) search over graph hyperparameters from the (expensive) FinBERT retraining step, which only has to be run once the graph configuration and confidence threshold have already been chosen.

The approach also has structural limitations. Its results are highly sensitive to the quality of the underlying embeddings: if the representation space does not accurately reflect semantic and sentiment similarity – for instance because the Approach 2 encoder is imperfect, or because it places posts of different sentiment close together – the graph will be misleading and propagation will fail. Graph construction also requires careful tuning: small changes in k or α can produce substantially different propagation outcomes, as shown by the hyperparameter search in Section 4.5.3. Finally, like Approach 3, Approach 4 still retrains a classifier on its selected pseudo-labels, and is therefore not immune to the same error-propagation risk if those pseudo-labels are systematically wrong rather than merely noisy; label spreading changes how pseudo-labels are generated and filtered, but does not remove this risk entirely.

4.6. Summary and Choice of Sentiment Model for Forecasting

The four approaches developed in this chapter trace a single, coherent progression from a generic, off-the-shelf sentiment model to a model specifically adapted to StockTwits data on SPY. Table 4.9 collects their headline results on the shared held-out test set.

Approach	Accuracy	F1 macro
1 – Zero-shot FinBERT	0.5780	0.5778
2 – Supervised fine-tuning	0.8008	0.7992
3 – Semi-supervised self-training	0.8073	0.8067
4 – Semi-supervised label spreading	0.8088	0.8081

Table 4.9: Headline test-set performance of all four approaches developed in this chapter, evaluated on the same held-out Bullish/Bearish test set ($n = 283,943$).

The largest gain by a wide margin comes from adapting FinBERT to the target data at all: fine-tuning on labelled StockTwits posts (Approach 2) improves accuracy by more than 22 percentage points over the zero-shot baseline (Approach 1), confirming that the central obstacle identified in Section 4.2 – the distribution shift between FinBERT’s original fine-tuning corpus and short, informal financial social media text – is real and substantial, and that a comparatively simple, single-stage fine-tuning step is sufficient to close most of it.

By contrast, the two semi-supervised extensions add only a marginal improvement on top of this already-adapted model: Approach 3 improves macro F1 by 0.0075 over Approach 2, and Approach 4 by 0.0089, despite each approach exploiting well over a million additional, pseudo-labelled posts. Crucially, this is not the result of an under-explored design space. Each approach was pushed further than a single default configuration would require: two structurally different pseudo-labelling philosophies were compared under matched conditions – one model-driven (confidence and neighbourhood agreement on a classifier’s own predictions), one structure-driven (diffusion over a similarity graph); the core assumption behind both, that higher-confidence pseudo-labels are also more often correct, was tested directly against withheld ground truth rather than taken on faith (Sections 4.4.3 and 4.5.4); each approach was retrained at stricter, quality-optimised thresholds to check whether more careful filtering translates into a better final classifier (Sections 4.4.5 and 4.5.5); and the possibility that scarce labelled data was masking a larger semi-supervised benefit was tested directly by repeating Approaches 2, 3, and 4 on shrinking fractions of the labelled training set.

Labelled fraction	Approach 2	Approach 3	Approach 4
1%	0.701	0.691	0.708
5%	0.741	0.741	0.749
10%	0.761	0.761	0.766
20%	0.776	0.778	0.780

Table 4.10: Test macro F1 for Approaches 2, 3, and 4 when trained on shrinking fractions of the labelled training data, evaluated on the same fixed test set used throughout this chapter ($n = 283,943$).

If the modest advantage of Approaches 3 and 4 over Approach 2 were an artefact of already having abundant labelled data, it should widen as the labelled fraction shrinks and pseudo-labelled data becomes comparatively more valuable. Table 4.10 shows the opposite: the three approaches remain close together at every fraction, with gaps of at most a few thousandths of a point except at 1%, where the ranking is in any case not stable enough to draw conclusions from a single labelled seed size. Taken together with the consistently marginal gains observed on the full dataset, this indicates that the ceiling on Bullish/Bearish classification performance in this setting is set by something other than the quantity of labelled or pseudo-labelled data – most plausibly the intrinsic difficulty and ambiguity of short, informal StockTwits posts and of the self-reported Bullish/Bearish tags used as ground truth, which no amount of additional pseudo-labelled data, however carefully filtered, can fully resolve.

On this evidence, Approach 2 is retained as the sentiment model carried forward into the forecasting pipeline described in the remainder of this thesis, rather than Approach 3 or Approach 4. This choice is not a claim that semi-supervised learning failed: both approaches deliver a small but genuine improvement, replicated under two different pseudo-labelling mechanisms and confirmed to be robust to threshold choice and to the amount of labelled data available. It is instead a judgement about where the added complexity is worth its cost. Approaches 3 and 4 each require an additional pseudo-labelling pipeline, a further round of hyperparameter search, and a second full FinBERT retraining on a substantially larger, partly machine-generated training set, for a gain of less than one accuracy point that is unlikely to survive, let alone be decision-relevant, once sentiment predictions are aggregated into the daily-level features used for forecasting in the next chapter. Approach 2 achieves comparable performance with a single, simpler, and more easily reproducible training procedure, and is therefore the more parsimonious choice for a component that is not itself the central object of this thesis, but an input to it. Approaches 3 and 4 are reported in full because they constitute a thorough and, to the best of this thesis’s knowledge, novel head-to-head comparison of two semi-supervised paradigms

on financial social media text under a matched embedding space and evaluation protocol (Section 3.3.3); they remain part of this thesis as a complementary methodological contribution, rather than as the source of the sentiment signal used downstream.

With the sentiment classification stage now settled, the central question this thesis was designed to answer moves to the foreground: does the sentiment signal extracted from financial social media text, using the Approach 2 classifier, add genuine predictive value for financial forecasting, beyond what purely financial features – including the market’s own volatility benchmark, the VIX – already capture? The remainder of this chapter turns to this question directly.

5 | Forecasting Framework: From Sentiment to Predictive Value

5.1. From Text to Signal

The first part of this thesis focused on building a reliable sentiment classifier. A FinBERT model, fine-tuned through progressive layer unfreezing, achieved a macro F1-score of 0.799 on the StockTwits SPY dataset. To exploit the large pool of unlabeled tweets, two semi-supervised approaches — confidence-based pseudo-labeling and label spreading — were also investigated. However, neither method produced meaningful improvements over the supervised baseline, even in low-data scenarios.

These findings naturally shifted the focus of the study. Rather than pursuing increasingly small gains in tweet-level classification performance, the remainder of the thesis investigates a more practically relevant question: can sentiment extracted from social media improve financial forecasting?

Addressing this question requires moving from the level of individual tweets to that of the market. Single sentiment predictions are inherently noisy and provide limited information in isolation; their value emerges only after being aggregated into time-varying indicators that describe the overall market mood. To this end, approximately 3.2 million tweet-level predictions were transformed into a daily sentiment feature set capturing not only the average market sentiment, but also its temporal evolution, the degree of agreement or disagreement among investors, and changes in market attention through message volume.

These sentiment features were then combined with a complementary set of market indicators, including momentum, realized volatility, and measures of market-wide risk, to form the input space for the forecasting experiments presented in this chapter.

5.2. Two Research Questions

To investigate the central research question of *whether sentiment extracted from financial social media provides predictive information beyond that already contained in financial*

time series, the analysis is divided into two complementary questions. The first concerns the **incremental predictive value** of sentiment. Specifically, does a model combining financial and sentiment features outperform an equivalent model trained solely on financial variables? This represents a more demanding criterion than simply showing that sentiment is correlated with future market behaviour. A statistically significant correlation does not imply that sentiment contributes new information, as the observed relationship may already be explained by variables such as trading volume or realized volatility. To address this, every forecasting experiment compares three models under the same evaluation framework: a financial-only baseline, a sentiment-only model, and a combined model. This comparison is performed across nine forecasting tasks, covering return, direction, and volatility at three prediction horizons (1 day, 3 days, and 1 week).

The second question concerns **when and under which market conditions** sentiment is informative. Even if the combined model shows little or no improvement on average, sentiment may still provide useful information in specific situations, such as periods of elevated market attention, strong disagreement among investors, or unusually high volatility. The exploratory analysis in Chapter 5.3 already pointed towards this possibility. Consequently, the objective is not only to determine whether sentiment improves forecasting overall, but also to identify the market regimes in which its contribution becomes meaningful. This motivates the robustness analyses presented in the second half of the chapter.

5.3. Preliminary Exploratory Data Analysis

Before constructing the candidate feature sets described in the following section, an exploratory analysis of the sentiment and price series was carried out, examining the correlation between sentiment and the forecasting targets, the temporal structure of the sentiment series, the redundancy among candidate features, and the stability of these relationships over the sample period. The findings summarized here motivate several of the modeling choices made later in this chapter.

5.3.1. Correlation with Forecasting Targets

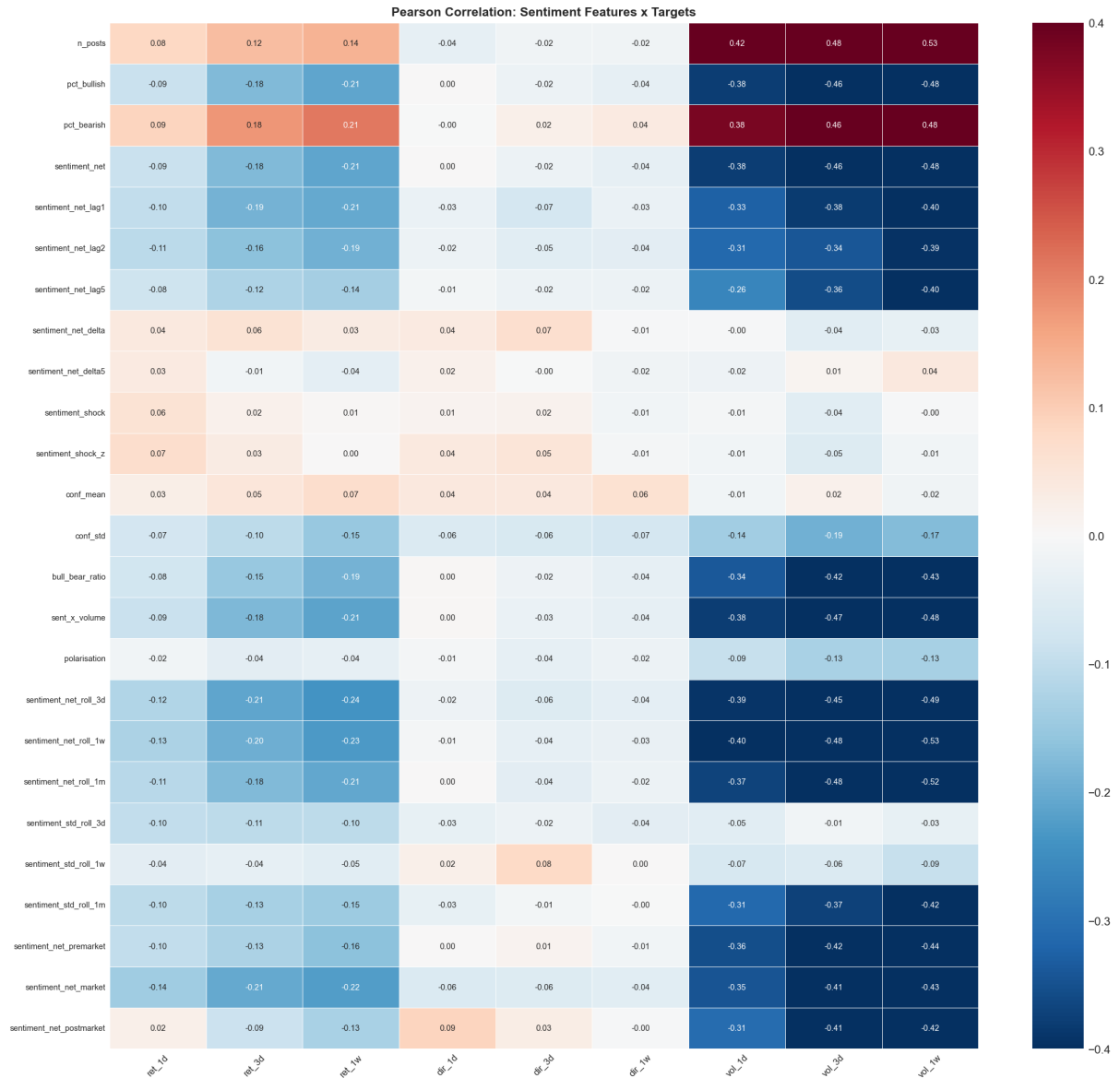


Figure 5.1: Pearson correlation between sentiment features and the nine forecasting targets.

The correlation analysis reveals a clear pattern across the three forecasting tasks. Sentiment exhibits only a weak relationship with future returns, almost no association with market direction, and a substantially stronger relationship with future volatility.

For returns, `sentiment_net` shows a modest but consistently negative correlation, increasing in magnitude from $r = -0.09$ at the one-day horizon to $r = -0.21$ after one week. Although relatively small, this pattern is consistent with a mild contrarian effect,

whereby periods of stronger retail optimism tend to be followed by weaker subsequent returns.

The strongest relationships emerge for volatility. Net sentiment is negatively correlated with future realized volatility ($r = -0.38$ to -0.48 across the three horizons), while message volume (`n_posts`) displays the largest correlation observed in the analysis, ranging from $r = 0.42$ to 0.53 . This finding suggests that the level of investor attention, independently of whether sentiment is bullish or bearish, contains useful information about subsequent market uncertainty. For this reason, message volume was retained as a forecasting feature rather than being treated solely as a control variable.

By contrast, the directional targets (`dir_*`) exhibit only negligible correlations with all sentiment features, providing an early indication that sentiment is likely to play a more limited role in directional forecasting than in volatility prediction.

The analysis also compares sentiment computed over different intraday windows. Somewhat unexpectedly, sentiment measured during market hours (`sentiment_net_market`) exhibits stronger correlations with both future returns ($r = -0.14$ to -0.22) and volatility ($r = -0.35$ to -0.43) than the pre-market measure ($r = -0.10$ to -0.16 and $r = -0.36$ to -0.44 , respectively). This contrasts with much of the existing literature, which typically focuses on pre-market sentiment because it precedes trading activity. Since the same pattern appears under both Pearson and Spearman correlation, it is unlikely to be driven by a small number of outliers. Consequently, as discussed in Section 5.4.1, both intraday windows were retained as candidate forecasting features.

5.3.2. Autocorrelation and Persistence

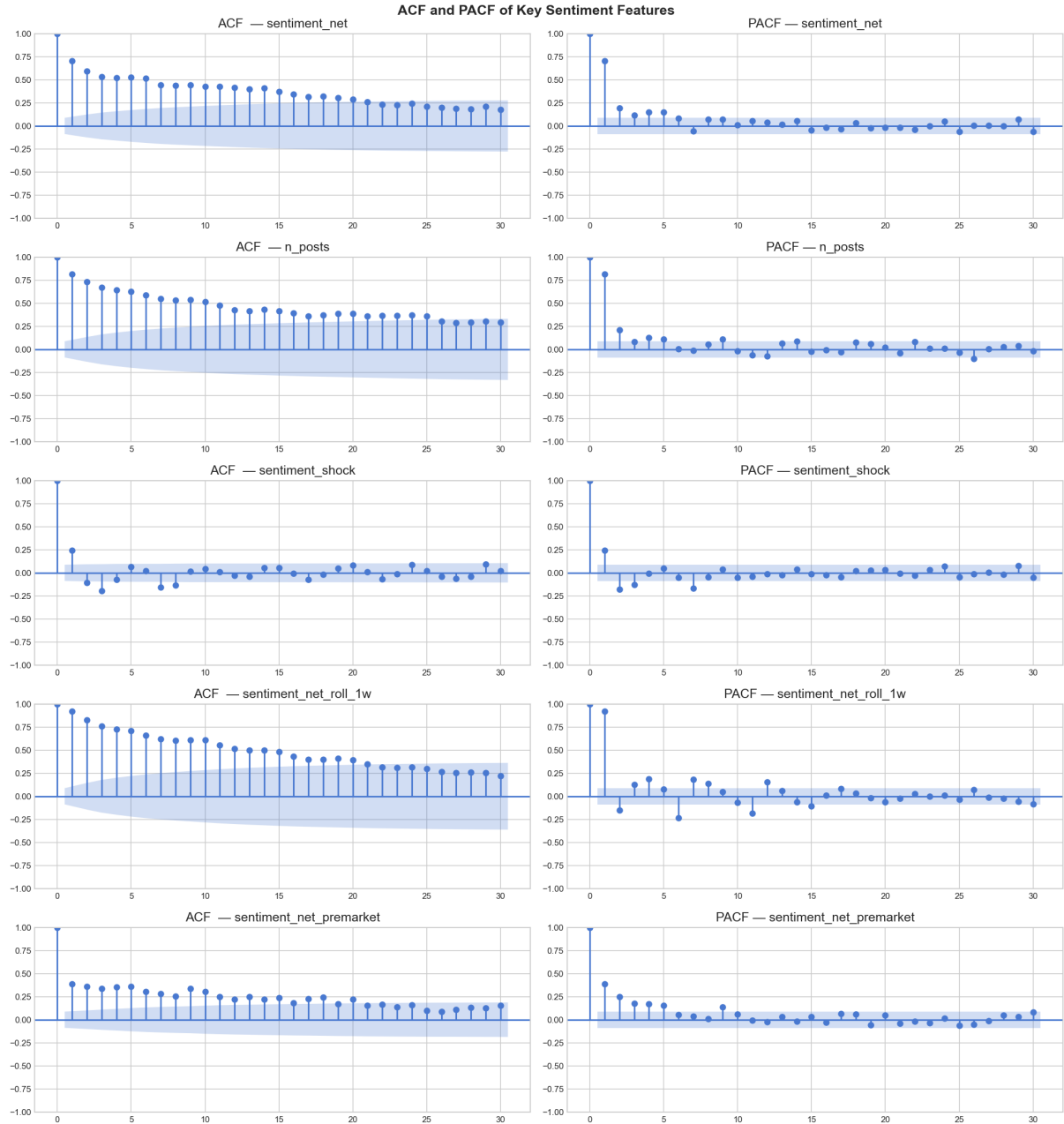


Figure 5.2: Autocorrelation (ACF) and partial autocorrelation (PACF) of key sentiment features.

The autocorrelation analysis shows that `sentiment_net` is moderately persistent over time. Its ACF decays gradually, while the PACF drops sharply after the first lag, a pattern characteristic of an AR(1)-like process. In practice, yesterday's sentiment contains useful information about today's sentiment, whereas observations further in the past contribute

little once the first lag is taken into account.

Message volume (`n_posts`) displays a different behaviour. Its autocorrelation is substantially more persistent, with the PACF remaining noticeable at the second lag. This suggests that periods of unusually high or low retail activity tend to last longer than changes in the overall sentiment expressed by investors.

By contrast, the sentiment shock feature, defined as the deviation of daily sentiment from its own 5-day rolling average, behaves almost like white noise. This indicates that it captures short-lived departures from the prevailing sentiment level without inheriting the persistence of the original series.

To complement the autocorrelation analysis, an Augmented Dickey–Fuller test was performed on all candidate sentiment features. The test indicated that `sentiment_net`, the bullish and bearish proportions, sentiment shock, polarisation, and the rolling dispersion measures are stationary. In contrast, message volume, the rolling sentiment averages (3-day, 1-week, and 1-month), the mean and dispersion of classifier confidence, and lagged sentiment are not. These results motivated the log transformation applied to message volume before model training (Section 5.4) and are consistent with the persistence observed in the original series.

5.3.3. Feature Redundancy

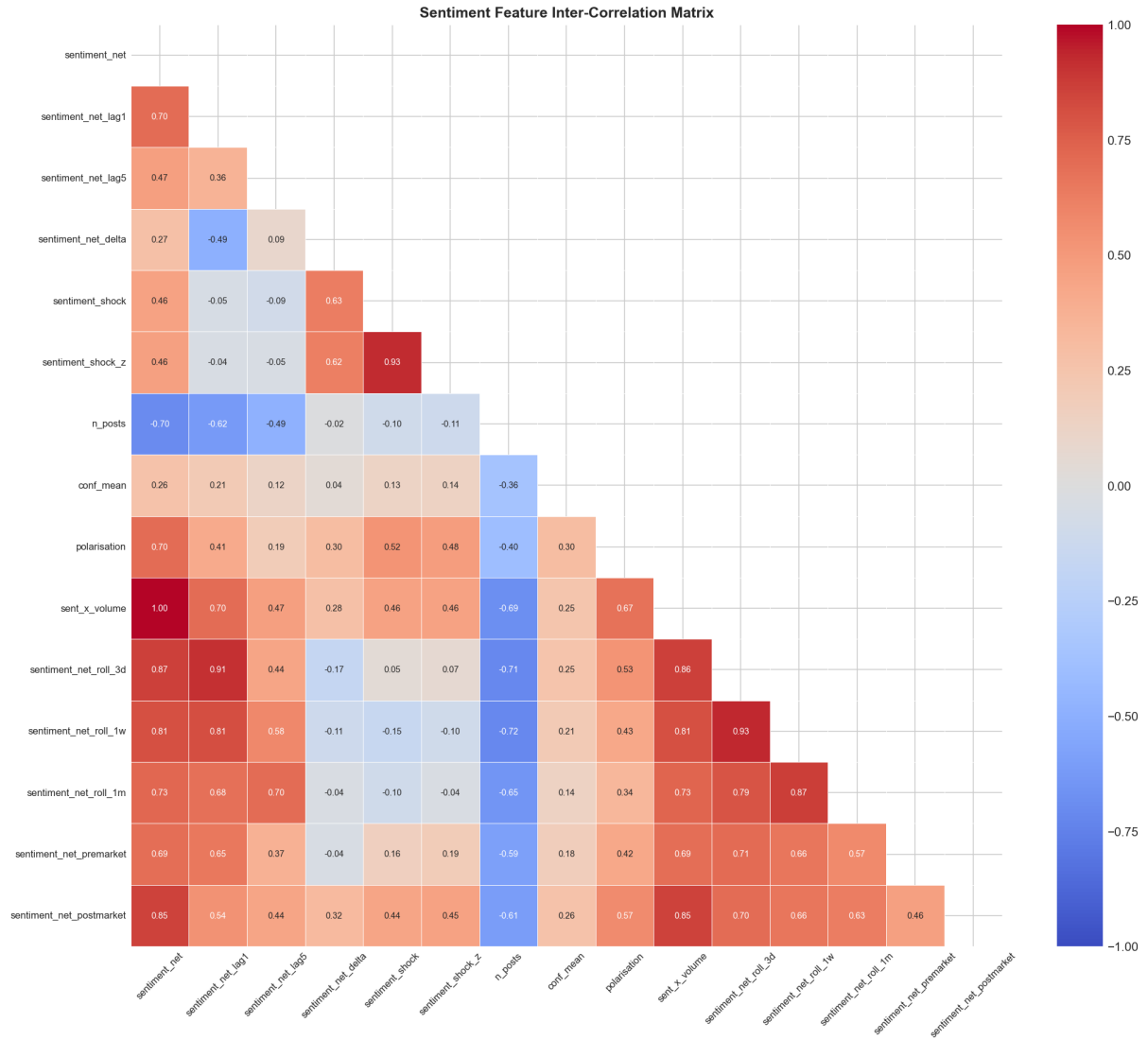


Figure 5.3: Pairwise correlation among candidate sentiment features.

The correlation matrix reveals substantial redundancy among several candidate sentiment features. Many variables are not independent pieces of information but different representations of the same underlying signal. For example, the interaction term `sent_x_volume` is almost perfectly correlated with `sentiment_net` ($r = 1.00$), while the bullish proportion, bearish proportion, and bull/bear ratio are all highly correlated because they are simple transformations of the same sentiment distribution. Similarly, the 3-day and 1-week rolling averages of sentiment exhibit a strong correlation ($r = 0.93$), as do the sentiment shock feature and its standardized (z-score) version.

Message volume stands out as an exception. Compared with the other sentiment vari-

ables, it displays considerably lower correlations ($|r| \leq 0.72$ with every other candidate), suggesting that it captures a different aspect of investor behaviour. Rather than reflecting sentiment itself, it measures the level of market attention, making it a potentially complementary source of information.

These observations indicated that retaining every engineered feature would introduce unnecessary redundancy into the forecasting model. This motivated the systematic feature selection procedure described in Section 5.4.1, where correlation analysis is combined with variance inflation factors (VIF) to identify a more compact and informative feature set.

5.3.4. Temporal Stability

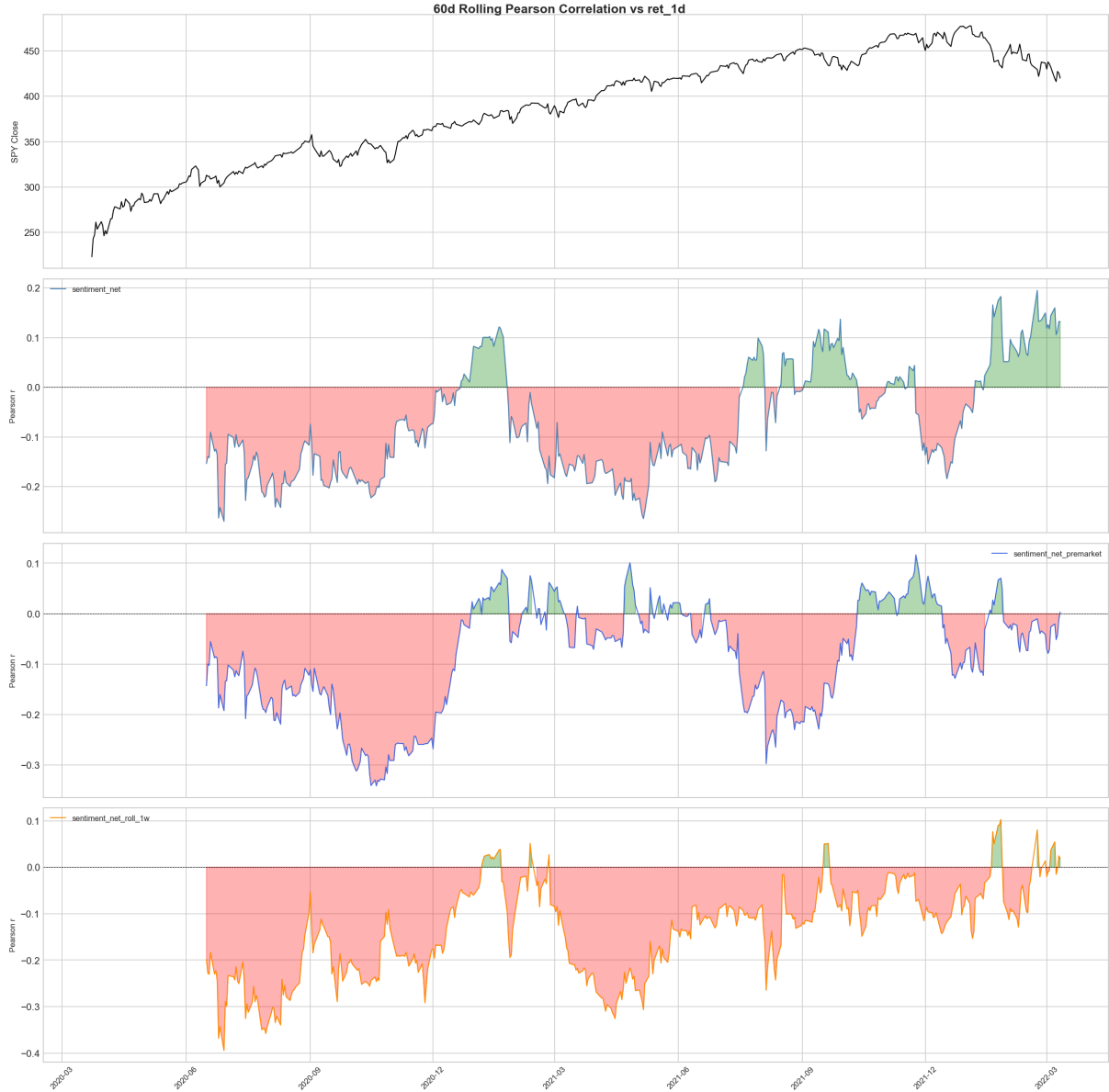


Figure 5.4: 60-day rolling Pearson correlation between sentiment and one-day-ahead return.

The relationship between sentiment and future returns varies considerably over time. Figure 5.4 shows the 60-day rolling Pearson correlation between `sentiment_net` and one-day-ahead returns. Instead of remaining stable throughout the sample, the correlation changes direction several times, ranging from values close to -0.25 to short periods of positive correlation. This suggests that the predictive content of sentiment is unlikely to be constant over time.

To investigate this further, the sample was divided into quartiles based on message volume, sentiment polarisation, and market-wide volatility. The contrarian relationship between sentiment and future returns becomes noticeably stronger during periods of high retail participation and high investor disagreement, reaching Pearson correlations as low as -0.40 in the highest polarisation quartile. This is consistent with previous work suggesting that disagreement among investors increases the informational content of social-media sentiment Cookson and Niessner, 2020.

A different pattern emerges during periods of elevated market volatility. Here, the correlation between sentiment and future returns becomes much weaker and, in some cases, even changes sign. One possible interpretation is that, during highly volatile markets, retail investors react to ongoing market movements instead of anticipating them, making sentiment more of a contemporaneous indicator than a leading one.

Overall, these results suggest that the usefulness of sentiment depends on the market environment. Its contribution is not constant across time or market conditions, providing a strong motivation for the regime-specific robustness analyses presented later in this chapter.

5.4. Feature Construction and Selection

Having established a reliable sentiment classifier, the next step is to transform its tweet-level predictions into a structured set of variables suitable for financial forecasting. This section first describes how the candidate sentiment and financial features were constructed, and then how they were reduced to a smaller set of informative, non-redundant predictors.

Two feature pools were built independently. The first consists of 25 sentiment-derived features computed from the FinBERT predictions, while the second contains 21 financial features extracted from the daily OHLCV data of SPY together with the VIX index. Although the two groups originate from different data sources, both were processed using the same feature selection pipeline to ensure a consistent comparison.

5.4.1. Selection Methodology

The complete feature space contains 46 candidate variables, while the forecasting dataset comprises only around 500 daily observations. Using all available features would therefore increase the risk of overfitting and multicollinearity. To reduce this risk, the same three-stage feature selection procedure was applied to both the financial and sentiment feature sets.

The first step evaluates the relevance of each feature using mutual information with the

nine forecasting targets (return, direction, and volatility at the three prediction horizons). Mutual information was preferred over simple correlation because it captures both linear and non-linear relationships. Since its estimation is stochastic and the dataset is relatively small, each score was averaged across five random seeds to improve stability.

Features were then ranked according to their average mutual information score. Starting from the highest-ranked feature, variables were selected greedily, provided their correlation with the already-selected features remained below 0.85. Features exceeding this threshold were not discarded, but stored as reserve candidates.

Finally, the selected feature set was refined using an iterative Variance Inflation Factor (VIF) analysis. Whenever a feature exceeded a VIF of 10, it was removed and replaced by the most relevant reserve feature from the same correlated group. This prevents an entire family of related features—for example, rolling sentiment averages—from being excluded simply because the first representative selected exhibited excessive multicollinearity.

The final procedure retained 14 of the 25 sentiment features and 17 of the 21 financial features. The resulting feature sets are presented in the sections below.

Selected Sentiment Features

- The average net sentiment over the trailing month, `sentiment_net_roll_1m` = $\frac{1}{21} \sum_{i=0}^{20} \text{sentiment_net}_{t-i}$, was the highest-ranked feature in the mutual information screening, capturing a slow-moving, structural retail mood rather than day-to-day noise.
- Same-day sentiment restricted to market-hours tweets (09:30–16:00 EST), `sentiment_net_market`, was found to correlate more strongly with forward returns and volatility than the pre-market window, despite the latter being the more common choice in the sentiment literature.
- Daily message volume, `n_posts`, used as a proxy for retail attention independent of tone, was the strongest single correlate of realized volatility in the exploratory correlation analysis.
- Sentiment lagged by two trading days, `sentiment_net_lag2` = `sentiment_net`_{*t*-2}, was retained because the autocorrelation analysis showed sentiment exhibits AR(1)-like persistence, meaning short lags carry information not fully captured by the rolling averages.
- The rolling standard deviation of daily net sentiment over the trailing month, `sentiment_std_roll_1m`, distinguishes sustained disagreement among retail participants from a temporary spike in polarisation.

- Pre-market sentiment (04:00–09:30 EST), `sentiment_net_premarket`, was retained alongside its market-hours counterpart as a theoretically motivated, ex-ante signal available before trading begins, even though it proved empirically weaker.
- The short-window rolling standard deviation of sentiment, `sentiment_std_roll_1w`, measures how much day-to-day sentiment fluctuates within a trading week, and is interpreted as a proxy for short-term disagreement among retail participants.
- The mean FinBERT confidence score across a day’s tweets, `conf_mean`, is included as a data-quality control, on the premise that low average confidence may indicate a day of mixed or ambiguous market narratives.
- Polarisation, `polarisation` = $|\text{sentiment_net}|$, measures how one-sided the crowd is on a given day, regardless of direction, distinguishing a strongly one-sided market mood from a genuinely balanced one.
- Sentiment shock, `sentiment_shock` = $\text{sentiment_net}_t - \text{MA}_5(\text{sentiment_net})_t$, isolates a sudden shift in tone from the slower-moving baseline already captured by the rolling-average features.
- The within-day dispersion of classifier confidence, `conf_std`, complements the mean confidence feature: a day with highly variable confidence may reflect a genuinely mixed market narrative, distinct from low average confidence alone.
- The one-week change in net sentiment, `sentiment_net_delta5` = $\text{sentiment_net}_t - \text{sentiment_net}_{t-5}$, captures medium-term sentiment momentum, distinct from both the 1-day change and the rolling-average level.
- The very short-window rolling dispersion of sentiment, `sentiment_std_roll_3d`, captures how much sentiment disagreement builds up or resolves within just a few trading days, the shortest dispersion window in the feature set.
- The one-day change in net sentiment, `sentiment_net_delta` = $\text{sentiment_net}_t - \text{sentiment_net}_{t-1}$, is the shortest-horizon momentum feature in the sentiment series, complementing the shock and 5-day change features above.

Selected Financial Features

- Long-window realized volatility, `price_vol_60d`, the rolling 60-day standard deviation of daily log-returns, was the highest-ranked financial feature in the mutual information screening, capturing the slow-moving volatility regime.

- The VIX closing level, `vix_level`, is the market's forward-looking, options-implied volatility expectation, external to SPY's own price history.
- Medium-window realized volatility, `price_vol_20d`, is a standard volatility-clustering proxy retained alongside the 60-day window.
- Ten-day price momentum, `price_ret_10d` = $\log(P_t/P_{t-10})$, consistently outranked both shorter and longer momentum horizons in relevance to the nine forecasting targets.
- Log-transformed trading volume, `log_volume` = $\log(1 + \text{Volume}_t)$, is strongly correlated with realized volatility; the log transform corrects the heavy right-skew of raw volume.
- The normalized intraday trading range, `price_range` = $(\text{High}_t - \text{Low}_t)/\text{Close}_t$, is a direct, same-day measure of realized price dispersion, complementary to close-to-close volatility measures.
- The price's distance from its 50-day moving average, a metric computed as `dist_sma_50` = $\text{Close}_t/\text{SMA}_{50}(\text{Close})_t - 1$, is a standard trend indicator, capturing whether the market is currently above or below its medium-term trend.
- The 5-day rolling average of the intraday range, `price_range_5d_avg`, smooths day-to-day noise in the raw range measure while retaining its short-term responsiveness.
- Twenty-day price momentum, `price_ret_20d`, was retained alongside the 5- and 10-day horizons to span short-, medium-, and longer-term momentum.
- Five-day price momentum, `price_ret_5d`, is the shortest multi-day momentum horizon retained, complementing the 1- and 2-day returns.
- The ratio of 10-day to 20-day realized volatility, `vol_ratio_10_20`, indicates whether short-term volatility is currently elevated or depressed relative to its medium-term level.
- The ratio of 5-day to 60-day realized volatility, `vol_ratio_5_60`, captures the same type of regime information at a wider time-scale contrast than the 10/20 ratio above.
- The overnight price gap, `gap_open` = $\log(\text{Open}_t/\text{Close}_{t-1})$, captures information that arrived after the previous session closed, which close-to-close return measures do not reflect.

- The one-day change in the VIX, $\text{vix_change_1d} = \log(\text{VIX}_t / \text{VIX}_{t-1})$, complements the VIX level with its short-term momentum.
- The most recent daily return, $\text{price_ret_1d} = \log(P_t / P_{t-1})$, is the baseline momentum feature at the shortest horizon, and the building block for the realized-volatility features above.
- Two-day price momentum, price_ret_2d , is a distinct short-horizon momentum signal from the 1-day return above.
- The one-day change in trading volume, volume_change_1d , captures sudden shifts in market activity not reflected in the level of $\log_volume$ alone.

5.5. Experimental Setup and Forecasting Methodology

5.5.1. Data Timeline and the Moment of Prediction

Before defining the forecasting targets, it is useful to clarify exactly which information is available at the time a prediction is made. Although the feature construction described above implicitly defines this timeline, making it explicit helps avoid ambiguity when interpreting the forecasting results.

All seventeen financial features are available at the close of trading on day t (16:00 EST). The daily open, high, low, close, trading volume, and VIX closing value are all fixed at that point, and every financial feature is computed exclusively from information available on day t or earlier.

The sentiment features follow a slightly different timeline. While `sentiment_net_premarket` and `sentiment_net_market` are also available by market close, several of the retained features—including `n_posts`, `sentiment_net_roll_1m`, `polarisation`, `sentiment_shock`, and the sentiment-change variables—are computed using the complete calendar-day collection of StockTwits posts, which extends until 24:00 EST.

Consequently, the combined forecasting model is evaluated assuming that predictions are generated at the end of calendar day t , after both the trading session and the day’s social media activity have concluded. This timing does not introduce look-ahead bias, since every feature is computed using information available before the prediction time, and every forecasting target is defined using observations that occur strictly afterwards.

The practical implication is that the models evaluated in this chapter represent an overnight forecasting setting. They are designed to predict future market behaviour once all information from day t has been observed, rather than to support intraday or pre-close

trading decisions. This distinction is important when interpreting the forecasting results and is revisited in the discussion (Section 5.8).

5.5.2. Forecasting Targets

The forecasting framework considers three prediction tasks based on daily SPY closing prices: future return, future direction, and future realized volatility. Each task is evaluated at three forecasting horizons (1 day, 3 days, and 1 week), resulting in nine targets overall.

All targets are defined using information that becomes available only after day t . Consequently, none of them overlaps with the information used to construct the financial and sentiment features described in the previous section. Figure 5.5 illustrates this separation between the information available when the prediction is made and the future prices used to compute each target.

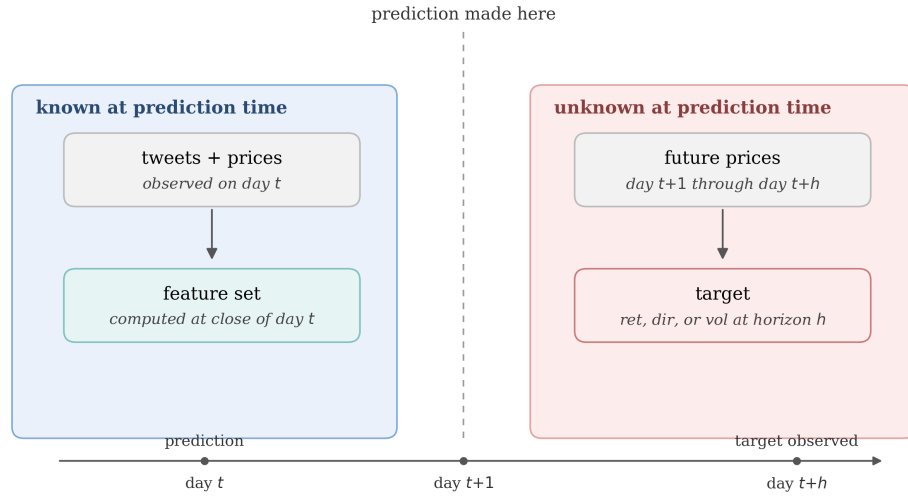


Figure 5.5: Timeline separating information available at the point of prediction (left) from the future prices used to compute each target (right).

Future return,

$$\text{ret}_h = \log \left(\frac{P_{t+h}}{P_t} \right),$$

is treated as a regression target. Future direction,

$$\text{dir}_h = \mathbb{1}[\text{ret}_h > 0],$$

is derived directly from the future return and framed as a binary classification task.

Future realized volatility, denoted by vol_h , is also modelled as a regression problem. At the one-day horizon, where only a single future return is available, volatility is approximated by the absolute return,

$$\left| \log \left(\frac{P_{t+1}}{P_t} \right) \right|.$$

For the three-day and one-week horizons, it is computed as the standard deviation of daily log-returns over the corresponding forward window.

These three targets capture different aspects of market behaviour. Return measures the magnitude and direction of future price movements, making it the quantity most directly related to trading performance. Direction focuses only on the sign of the movement and therefore represents a simpler prediction task, since the exact size of the move does not need to be estimated. Volatility, instead, measures the variability of future returns regardless of direction, making it more closely related to market risk than to price forecasting. Evaluating all three tasks allows the contribution of sentiment to be assessed across different forecasting objectives, instead of assuming in advance that it should benefit only one of them.

Task	Horizon	Mean	Std. dev.	% positive
Return	1 day	0.0013	0.0125	–
	3 days	0.0035	0.0205	–
	1 week	0.0055	0.0244	–
Direction	1 day	–	–	56.8
	3 days	–	–	60.2
	1 week	–	–	61.6
Volatility	1 day	0.0088	0.0089	–
	3 days	0.0096	0.0075	–
	1 week	0.0101	0.0067	–

Table 5.1: Descriptive statistics of the nine forecasting targets. Mean and standard deviation are reported for regression targets; % positive is reported for direction (share of days with $\text{dir}_h = 1$).

Table ?? highlights two characteristics of the forecasting targets that are useful when interpreting the results.

First, the proportion of positive returns increases with the forecasting horizon, from 56.8% at one day to 61.6% after one week. This reflects the positive drift of the equity market during the sample period and implies that always predicting the majority class provides a stronger baseline than random guessing, particularly at longer horizons.

Second, all three volatility targets exhibit pronounced right skewness (3.15, 1.78, and 1.83 for the one-day, three-day, and one-week horizons, respectively). A relatively small number of observations therefore account for a disproportionate share of realized volatility. For this reason, the robustness analysis in Section 5.7 reports mean absolute error alongside R^2 , since the latter is more sensitive to extreme observations.

5.5.3. Model Families and Feature Sets

Throughout this chapter, model performance is evaluated against two different baselines, each serving a different purpose.

The first is the **naïve baseline**: the training-set mean for the regression tasks and the majority class for direction. Its only purpose is to verify that a model has learned something beyond a trivial prediction.

The second is the **financial-only model**, which is the main reference point of this thesis. The central research question is whether sentiment provides information beyond what is already contained in financial variables. To answer this, every forecasting task compares three feature sets under identical experimental conditions: a financial-only model using the 17 selected financial features, a sentiment-only model using the 14 sentiment features, and a combined model using both. Throughout this chapter, sentiment is considered to add predictive value only if the combined model outperforms the financial-only baseline.

Each feature set was evaluated using three learning algorithms chosen to represent different modelling assumptions, rather than to identify a single "best" forecasting model.

- **Ridge regression** (for return and volatility) and **logistic regression** (for direction), both with regularization strength $\alpha = 1.0$. These linear models assume an additive relationship between predictors and target, providing a simple and interpretable benchmark for assessing the benefit of more flexible methods.
- **Random Forest**, using 300 trees with a maximum depth of 5. By combining multiple decision trees, Random Forest captures non-linear relationships and feature interactions without requiring them to be specified explicitly. The maximum depth was intentionally limited to reduce overfitting given the relatively small training sets in the early walk-forward folds.
- **XGBoost**, using 200 trees, a maximum depth of 3, and a learning rate of 0.05. Unlike Random Forest, gradient boosting builds trees sequentially, with each new tree correcting the errors made by the current ensemble. Although this often leads to better predictive performance on tabular datasets, it also increases the risk of

overfitting. Shallower trees and a conservative learning rate were therefore adopted to improve generalization.

The naive baseline is reported alongside every model and feature set throughout this chapter. As shown later in Section 5.7.5, surpassing this baseline should not be taken for granted, since several forecasting tasks remain challenging even for the more sophisticated models.

5.5.4. Evaluation Protocol

Model performance was evaluated using a walk-forward cross-validation scheme, which is appropriate for time-ordered financial data. At every iteration, each model is trained only on observations available before the test period, ensuring that no future information is used during training.

An initial implementation used a standard five-fold expanding-window split with no minimum training size. However, the first fold contained only around one sixth of the available observations, leading to highly unstable estimates, particularly for the tree-based models. The resulting variability reflected the limited amount of training data more than genuine forecasting performance.

To obtain more reliable estimates, a warm-up period covering the first 50% of the sample was introduced before the first evaluation fold. After this initial training period, five sequential, non-overlapping test folds were evaluated. At each step, the training set expands to include all observations available up to that point, while the following block of observations is used for testing.

Fold	Train size	Test size	Test period
1	249	49	Mar 18 – May 26, 2021
2	298	49	May 27 – Aug 5, 2021
3	347	49	Aug 6 – Oct 14, 2021
4	396	49	Oct 15 – Dec 23, 2021
5	445	53	Dec 27, 2021 – Mar 11, 2022

Table 5.2: Walk-forward fold boundaries (illustrated for the 498-observation one-day-horizon targets; the total number of observations, and consequently the exact fold boundaries, varies slightly by target due to the trailing observations lost when computing longer-horizon targets).

This evaluation protocol results in

$$9 \text{ targets} \times 3 \text{ feature sets} \times 4 \text{ models} \times 5 \text{ folds} = 540$$

individual model fits, from which all results reported in Section 5.6 are obtained.

Regression models were evaluated using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE). Classification models were evaluated using accuracy, F1-score, and the area under the ROC curve (AUC).

Unless otherwise stated, R^2 and AUC are reported as the primary performance metrics for regression and classification, respectively. Both are independent of the scale of the target variable and allow meaningful comparisons across forecasting horizons. In addition, R^2 has a natural interpretation relative to zero, making it straightforward to assess whether a regression model improves upon the naive baseline.

5.6. Main Results

Table 5.3 summarizes the main forecasting results. For each of the nine targets, it reports the primary evaluation metric (R^2 for return and volatility, AUC for direction) obtained by the naive baseline and by the best-performing model within each feature set, averaged across the five walk-forward test folds.

For every target, the strongest result among the three learning algorithms is reported. The goal is not to compare individual algorithms, but to assess the predictive potential of each feature set under the most favourable modelling approach. The final two columns show the performance difference between the combined and financial-only models, indicating whether adding sentiment improves over the financial baseline.

Table 5.3: Comparison of the naive baseline, financial-only, sentiment-only, and combined models across all nine forecasting targets. Metric is R^2 for return and volatility, AUC for direction. Δ is combined minus financial-only; a checkmark indicates the combined model improves on the financial-only baseline.

Task	Horizon	Naive	Fin.-only	Sent.-only	Combined	Δ	Improves
Return (R^2)	1 day	-0.042	-0.042	-0.093	-0.049	-0.008	✗
	3 days	-0.101	-0.072	-0.100	-0.084	-0.012	✗
	1 week	-0.198	-0.318	-0.280	-0.311	+0.008	✓
Direction (AUC)	1 day	0.500	0.504	0.571	0.548	+0.045	✓
	3 days	0.500	0.625	0.566	0.611	-0.014	✗
	1 week	0.500	0.666	0.594	0.628	-0.038	✗
Volatility (R^2)	1 day	-0.603	0.048	-0.152	0.125	+0.077	✓
	3 days	-1.002	0.051	-0.317	-0.254	-0.305	✗
	1 week	-1.500	-0.266	-0.533	-0.614	-0.347	✗

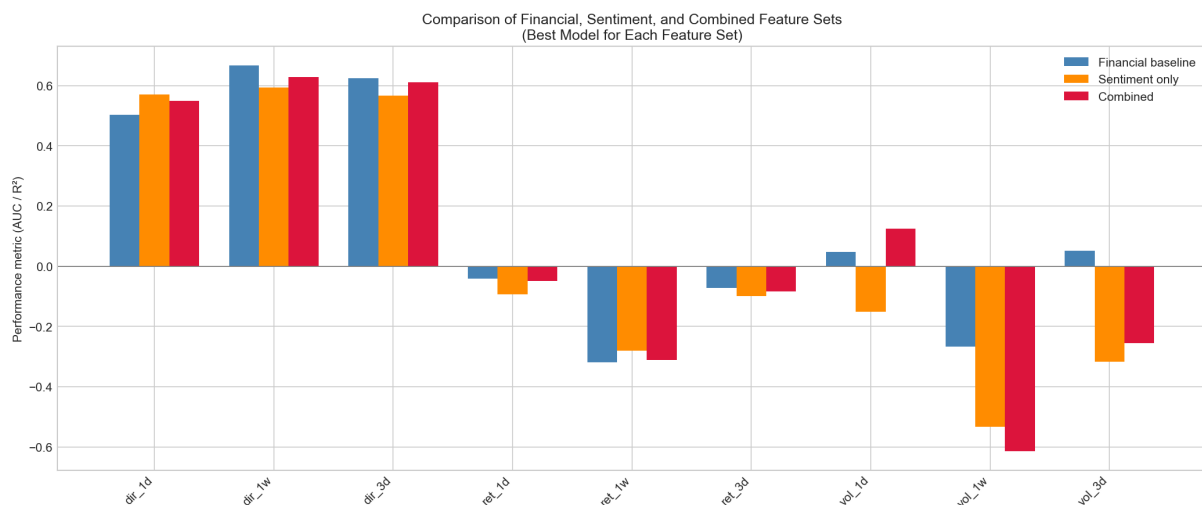


Figure 5.6: Comparison of financial-only, sentiment-only, and combined models across all nine forecasting targets, best-performing model per feature set. Positive values indicate the primary metric exceeds zero (uninformative for regression targets; better than chance for AUC).

Return

Return is the most difficult forecasting task considered in this chapter. Across all three horizons, none of the feature sets produces a meaningful improvement over the naive baseline.

At the one-day horizon, the financial-only model achieves an R^2 of -0.042 , identical to the naive benchmark. Adding sentiment does not improve performance, with the combined model reaching -0.049 , while the sentiment-only model performs substantially worse.

The three-day horizon shows a similar picture. Although the financial-only and combined models achieve slightly higher R^2 values than the naive baseline, the improvements are extremely small and unlikely to be practically meaningful. The sentiment-only model remains close to the naive benchmark.

At the one-week horizon, all three feature sets perform worse than the naive baseline. While the combined model exceeds the financial-only model by 0.008 R^2 points, both models remain well below the naive benchmark. In other words, sentiment slightly improves an already poor forecasting model, but not enough to produce a genuinely useful predictor.

Overall, these results suggest that future returns are extremely difficult to predict from either historical market variables or social-media sentiment. Within the experimental setting considered here, neither source of information provides consistent predictive power.

Direction

Unlike return forecasting, all three feature sets outperform the naive baseline ($AUC = 0.500$) at every prediction horizon, indicating that the data contain some useful directional information. Most of this predictive power comes from the financial features: the financial-only model improves steadily from an AUC of 0.504 at the one-day horizon to 0.666 after one week, suggesting that historical price dynamics become increasingly informative as the forecasting horizon lengthens.

Adding sentiment improves performance only at the one-day horizon, where the combined model reaches an AUC of 0.548 compared with 0.504 for the financial-only model ($\Delta = +0.045$). At the three-day and one-week horizons, however, the financial-only model remains the best-performing configuration, and incorporating sentiment leads to a small decrease in performance.

An interesting result emerges at the one-day horizon, where the sentiment-only model achieves the highest AUC (0.571), outperforming even the combined model. This suggests that sentiment contains short-term directional information that overlaps, at least in part, with the information already captured by financial variables. Since this effect disappears at longer horizons, it is treated as an isolated finding rather than as evidence that sentiment consistently improves directional forecasting (see Section 5.8).

Volatility

Volatility is the only forecasting task for which every feature set consistently outperforms the naive baseline. This is partly explained by the weakness of the naive predictor, whose R^2 falls to -1.500 at the one-week horizon because a constant mean prediction cannot capture the strong persistence typically observed in realized volatility.

Among the three feature sets, the combined model outperforms the financial-only baseline only at the one-day horizon, where R^2 increases from 0.048 to 0.125 ($\Delta = +0.077$). This represents the largest improvement associated with sentiment across all nine forecasting tasks.

The picture changes at longer horizons. At both three days and one week, the financial-only model clearly outperforms both the sentiment-only and combined models. In these settings, adding sentiment consistently reduces predictive performance, with performance drops of 0.305 and 0.347 R^2 points, respectively.

Overall, these results suggest that sentiment provides useful information for very short-term volatility forecasting, but its contribution fades rapidly as the prediction horizon increases.

Summary

Across the nine forecasting tasks, the combined model outperforms the financial-only baseline in only three cases: one-week return, one-day direction, and one-day volatility. The improvement in one-week return is not considered meaningful because neither model surpasses the naive baseline. The one-day direction result is more encouraging, but it is not replicated at longer horizons.

The clearest finding is therefore one-day volatility. Here, the combined model achieves the largest improvement over the financial-only baseline, and does so on a task where the financial model already provides substantial predictive power. This suggests that sentiment contributes genuinely complementary information, rather than simply improving an otherwise weak predictor.

Before examining this result further, it is important to understand how sensitive it is to the choice of financial baseline. The following section revisits the analysis after modifying the baseline to include one of the strongest available predictors of market volatility: the VIX.

5.6.1. The Role of VIX in the Financial Baseline

An earlier version of the analysis did not include the VIX level and its one-day change among the financial features. Revisiting this comparison is informative because it shows how sensitive the apparent contribution of sentiment is to the strength of the financial baseline itself.

Target	Baseline version	Financial-only	Combined	Δ
Volatility, 3 days	Without VIX	-0.318	-0.132	+0.107
	With VIX	0.051	-0.254	-0.305
Volatility, 1 week	Without VIX	-0.837	-0.614	+0.209
	With VIX	-0.266	-0.614	-0.347

Table 5.4: Effect of adding VIX to the financial baseline, volatility targets, best model per feature set.

Without VIX, the combined model appears to provide substantial improvements over the financial-only model for both the three-day and one-week volatility targets. Once VIX is included, however, the picture changes completely.

The financial-only model improves dramatically, particularly at the three-day horizon, where its R^2 increases from a strongly negative value to a positive one. As a result,

the apparent advantage of the combined model disappears entirely and, at the one-week horizon, even reverses.

This comparison suggests that the improvements observed in the earlier analysis were largely driven by an incomplete financial baseline rather than by genuinely new information contained in the sentiment features. Once one of the strongest predictors of future volatility is included, sentiment no longer adds value at the medium-term horizons.

The one-day volatility result is the only exception. Unlike the longer horizons, the combined model continues to outperform the financial-only baseline even after VIX is included, making it the most robust piece of evidence for incremental predictive value found in this study.

5.6.2. Conditional Analysis

The results above establish that the combined model’s advantage on one-day-ahead volatility holds on average, but do not indicate whether this advantage is uniform across market conditions or concentrated in specific regimes. Following the hypothesis, common in the sentiment literature, that social-media-derived signals may be more informative under conditions of high attention or high disagreement, out-of-fold predictions from both the financial-only and combined models were split into quartiles along two conditioning variables — sentiment polarisation and daily message volume — and the gain in R^2 within each quartile was computed.

Quartile	n	ΔR^2 (polarisation)	ΔR^2 (message volume)
Q1 (lowest)	63	−0.025	+0.378
Q2	62	+0.058	+0.015
Q3	62	−0.020	+0.015
Q4 (highest)	62	+0.143	−0.025

Table 5.5: Conditional R^2 gain (combined minus financial-only) on one-day-ahead volatility, by quartile. Each quartile contains approximately 62 observations.

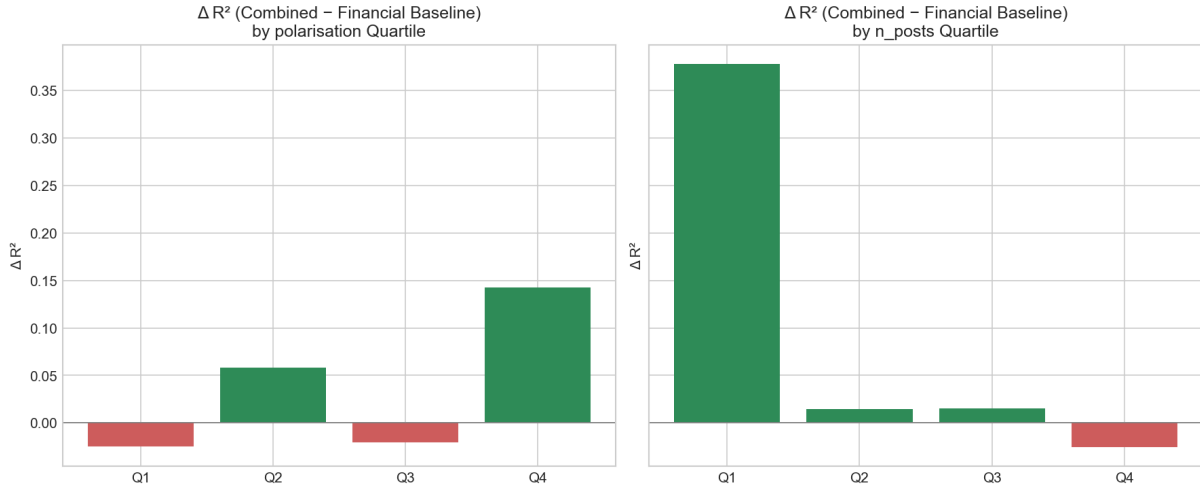


Figure 5.7: Conditional R^2 gain (combined minus financial-only) on one-day-ahead volatility, by quartile of sentiment polarisation (left) and daily message volume (right).

Conditioning on sentiment polarisation, the largest gain appears in the highest-polarisation quartile ($\Delta R^2 = +0.143$), partially consistent with the hypothesis that a more sharply divided retail crowd carries more useful information. Conditioning on message volume produces a different, and at first counter-intuitive, pattern: the largest gain by a wide margin appears in the *lowest*-volume quartile ($\Delta R^2 = +0.378$), not the highest. Inspection of the financial-only model's own performance within this quartile suggests an explanation: its R^2 there is -0.526 , its worst performance across all four quartiles, indicating that low-message-volume days are also the days on which the financial-only model struggles most on its own. The sentiment features appear to compensate specifically where the financial baseline is weakest, rather than adding value uniformly, or specifically where retail attention itself is highest. Given the small sample size within each quartile, this pattern is treated as indicative rather than conclusive, and is further examined for sensitivity to individual extreme observations in the following section.

5.7. Robustness Checks

The one-day-ahead volatility result remains after including the VIX in the financial baseline, but this alone is not sufficient to conclude that the effect is robust. The analyses presented in this section examine the result from several complementary perspectives. Specifically, they investigate whether the improvement is consistent across time, which sentiment features are responsible for it, how sensitive it is to extreme market events, whether it extends to a binary risk-management setting, and whether the same pattern can also be observed when forecasting the VIX itself.

5.7.1. Fold-by-Fold Decomposition

The results reported in Table 5.3 are averages across the five walk-forward test folds. While useful as an overall summary, this aggregation may hide whether the improvement is consistent throughout the sample or driven by a small number of periods. The table below therefore reports the one-day volatility performance of the financial-only and combined models separately for each test fold, using the Random Forest model.

Fold	Test period	Financial-only	Combined	Δ
1	Mar 18 – May 26, 2021	0.163	0.181	+0.018
2	May 27 – Aug 5, 2021	−0.410	0.059	+0.469
3	Aug 6 – Oct 14, 2021	0.078	−0.028	−0.106
4	Oct 15 – Dec 23, 2021	0.353	0.356	+0.004
5	Dec 27, 2021 – Mar 11, 2022	0.058	0.056	−0.002

Table 5.6: One-day-ahead volatility, R^2 by test fold. Δ is combined minus financial-only.

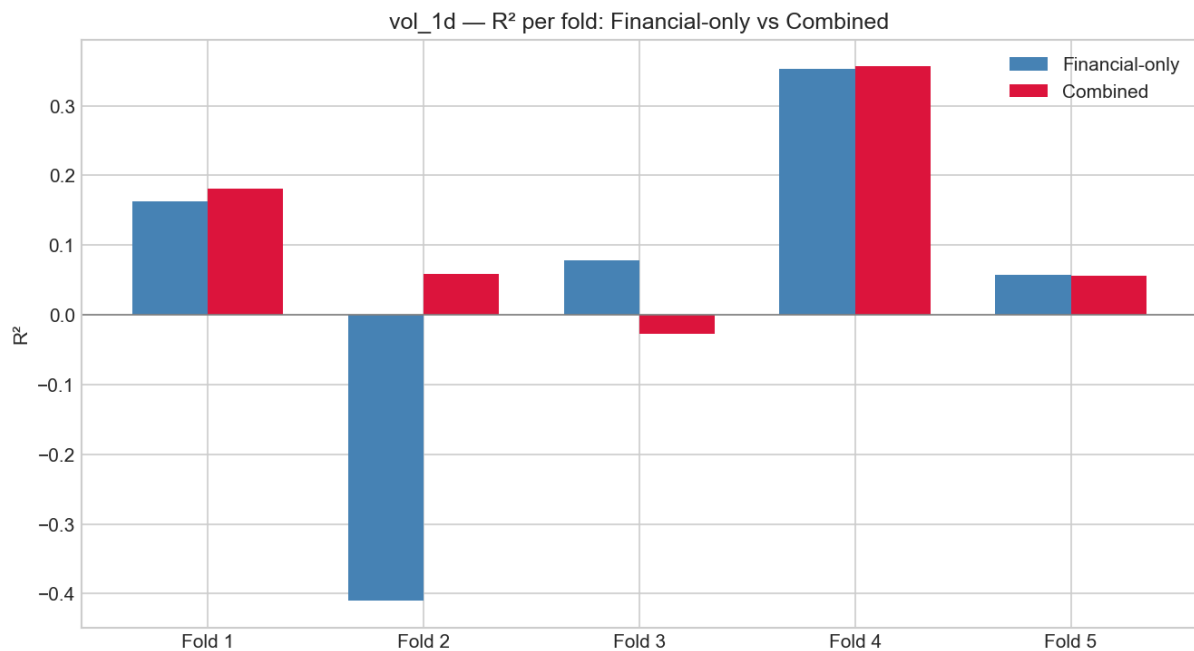


Figure 5.8: One-day-ahead volatility, R^2 by test fold, financial-only versus combined.

The improvement observed in the aggregate results is largely attributable to a single test fold. In Fold 2, the financial-only model records its weakest performance of the entire evaluation ($R^2 = -0.410$), whereas the combined model still achieves a positive R^2 of 0.059. This corresponds to an improvement of 0.469 R^2 points, exceeding the combined

gain observed across the other four folds. Outside this period, the two models perform very similarly, with differences generally within $\pm 0.02 R^2$ points and occasionally slightly favouring the financial-only model.

Fold 2 (May 27 – August 5, 2021) coincides with the market correction associated with concerns over the spread of the COVID-19 Delta variant, culminating in the sharp equity sell-off of July 19, 2021. One possible explanation is that this episode was driven by rapidly evolving news, which may not have been fully captured by financial features based solely on historical prices, trading volume, and the VIX. In contrast, sentiment extracted from real-time social-media discussions may have reflected these concerns earlier. Although this interpretation cannot be verified directly, it is consistent with the observation that the benefit of sentiment is concentrated in this specific period rather than being distributed uniformly across the sample.

5.7.2. Ablation Study

The feature importance analysis of the Random Forest model identified `sentiment_std_roll_3d`, the three-day rolling dispersion of sentiment, as the most important sentiment-derived predictor for one-day-ahead volatility. However, feature importance measures the contribution of a variable within a model that already contains all other predictors, including correlated ones, and therefore does not necessarily indicate how informative that feature would be on its own.

To better understand the contribution of individual sentiment variables, an ablation study was performed. Starting from the financial-only baseline, sentiment features were added one at a time and the resulting performance was compared with that of the full combined model containing all fourteen sentiment features.

Configuration	Mean R^2	Std. dev.	Δ vs. financial-only
Financial-only	0.048	0.282	–
+ message volume (<code>n_posts</code>)	0.101	0.184	+0.052
+ sentiment level (<code>sentiment_net_roll_1m</code>)	0.063	0.251	+0.015
+ sentiment dispersion (<code>sentiment_std_roll_3d</code>)	0.062	0.238	+0.014
+ all sentiment features (combined)	0.125	0.149	+0.077

Table 5.7: Ablation study, one-day-ahead volatility. Mean and standard deviation of R^2 over the five test folds.

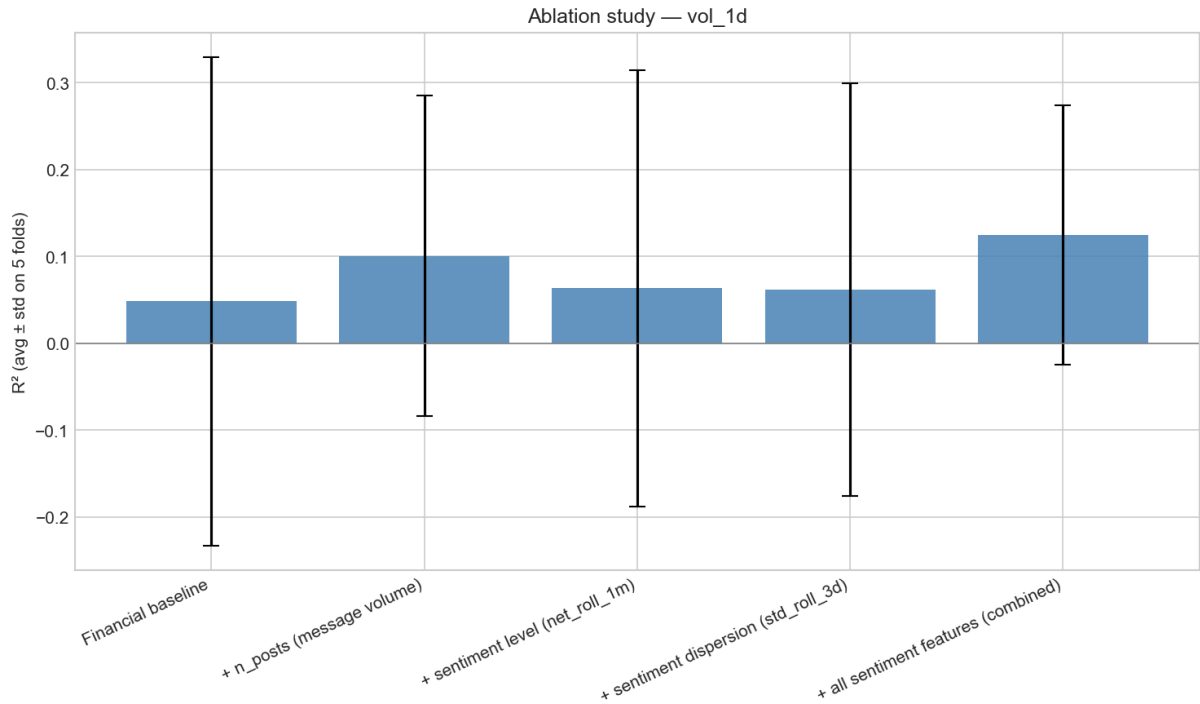


Figure 5.9: Ablation study, one-day-ahead volatility. Mean R^2 with error bars showing ± 1 standard deviation across the five test folds.

The results tell a different story from the feature importance ranking. Message volume (`n_posts`) emerges as the single most informative sentiment feature, increasing the mean R^2 from 0.048 to 0.101 when added to the financial baseline. By comparison, adding either the long-term sentiment level (`sentiment_net_rol_1m`) or the sentiment dispersion feature (`sentiment_std_rol_3d`) leads to much smaller improvements, raising R^2 by only 0.015 and 0.014, respectively.

This finding is consistent with the fold-by-fold analysis presented earlier. Message volume alone recovers much of the performance loss experienced by the financial-only model during the Delta-variant period, suggesting that market attention is the primary source of the short-term predictive gain.

At the same time, the best overall performance is achieved only when all sentiment features are used together. The combined model not only reaches the highest average R^2 , but also shows substantially lower variability across folds. Its standard deviation (0.149) is approximately half that of the financial-only model (0.282), indicating that the full sentiment feature set contributes not only to predictive performance but also to greater robustness across different market conditions.

5.7.3. Outlier Sensitivity

The conditional analysis (Section 5.6.2) showed the largest improvement of the combined model over the financial-only baseline in the lowest quartile of daily message volume ($\Delta R^2 = +0.38$), although this result is based on only about 63 observations. To assess whether the improvement was driven by a small number of unusually volatile days, the analysis was repeated after progressively removing the observations with the highest realized volatility within this quartile and re-evaluating the difference between the two models. Since repeated removal reduces the variance of the target and makes R^2 increasingly unstable, mean absolute error (MAE) was used for this robustness check instead.

The difference in MAE between the financial-only and combined models remained remarkably stable, ranging from 0.00054 to 0.00068 across all ten iterations, with no evidence that the improvement disappeared as the most extreme observations were removed. Moreover, only one of the five most volatile days belongs to the Delta-variant period discussed in Section 5.7.1; the other four are spread across the rest of the sample. This suggests that the conditional gain is not driven by a handful of extreme events and is likely distinct from the fold-specific effect observed during the Delta-variant episode.

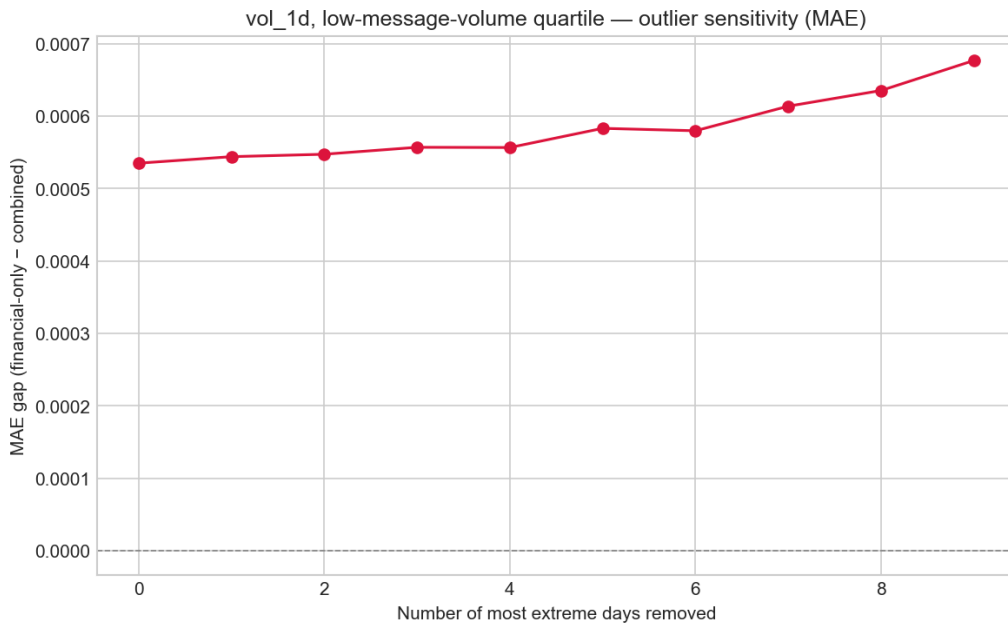


Figure 5.10: Sensitivity of the financial-only/combined MAE gap to the progressive removal of the most extreme-volatility days in the lowest-message-volume quartile.

5.7.4. Risk-Management Reframing

To assess whether the improvement observed for continuous volatility forecasting also has practical value for risk management, the problem was reformulated as a binary classification task. Instead of predicting the exact level of future volatility, the objective becomes identifying whether the next trading day belongs to the top 25% of realized one-day volatility.

The classification threshold (the 75th percentile of realized volatility) was computed separately within each training fold to prevent information from the test period influencing the target definition. For each feature set, the best-performing model was selected based on AUC, the primary classification metric used throughout this chapter. F1-score and Precision@top25% are also reported to provide a more complete evaluation.

Feature set	Model	AUC	F1	Precision@top25%
Financial-only	Logistic Regression	0.759	0.263	0.331
Sentiment-only	Logistic Regression	0.700	0.114	0.299
Combined	Random Forest	0.773	0.194	0.317

Table 5.8: One-day-ahead high-volatility classification (top 25%). Single model per feature set, selected by AUC, averaged over five folds.

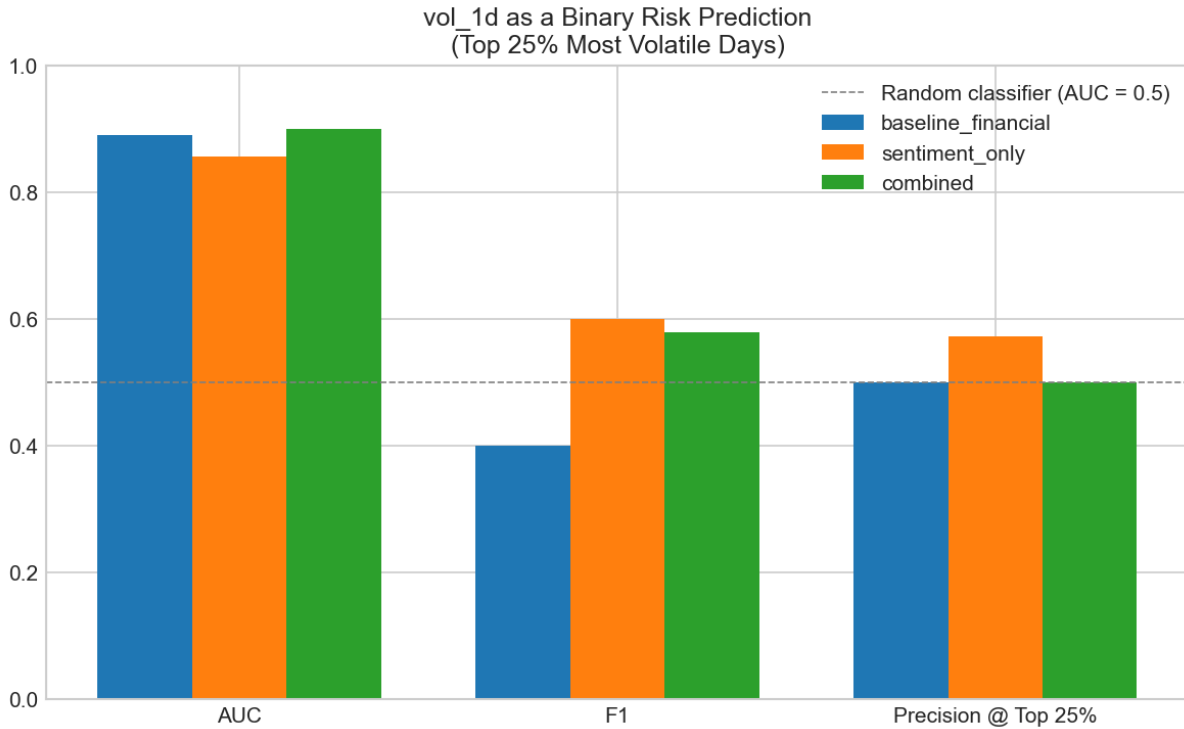


Figure 5.11: One-day-ahead high-volatility classification (top 25%): AUC, F1, and precision@top25%, best model per feature set.

The combined model achieves a slightly higher AUC than the financial-only model (0.773 versus 0.759), consistent with the improvement previously observed for the regression task. However, this advantage does not extend to the other evaluation metrics.

For both F1-score and Precision@top25%, the financial-only model performs best, achieving 0.263 and 0.331, compared with 0.194 and 0.317 for the combined model. Precision@top25% is particularly relevant from a risk-management perspective, as it measures how often the observations identified as highest risk are actually among the most volatile days.

These metrics should also be interpreted with some caution. Depending on the fold, the test set contains only four to nine positive examples (high-volatility days), making F1-score and precision substantially more variable than the regression metrics presented earlier.

Overall, the results suggest that the benefit of sentiment observed for continuous volatility forecasting does not translate into a clear advantage for binary risk classification. While sentiment slightly improves the overall ranking of observations, as reflected by AUC, it does not improve the identification of the highest-risk days. This suggests that sentiment is more informative about the expected magnitude of future volatility than about distinguishing the most extreme volatility events.

5.7.5. Transfer to VIX Forecasting

As a final robustness check, the analysis was repeated using the VIX itself as the forecasting target. Unlike realized SPY volatility, the VIX is an options-implied measure of expected market volatility and therefore represents a related, but distinct, prediction task. If the information captured by social-media sentiment reflects broader expectations about future market risk, it should also improve forecasts of future VIX movements.

To test this, a new target was defined as the one-day-ahead log-change in the VIX closing value, following the same forward-looking, no-lookahead framework adopted throughout this thesis. The financial feature set continued to include the current VIX level and its one-day change, together with the remaining financial variables.

Feature set	R^2
Naive (predicts the training-set mean)	−0.008
Financial-only	−0.094
Sentiment-only	−0.052
Combined	−0.060

Table 5.9: One-day-ahead VIX log-change, R^2 , best model per feature set.

None of the feature sets outperforms the naive baseline. Although the combined model performs slightly better than the financial-only model ($\Delta = +0.034$), both remain below the naive predictor, indicating that this improvement is not associated with meaningful predictive skill.

A fold-by-fold analysis (not reported in full) shows a pattern similar to that observed for realized volatility. The combined model performs better than the financial-only baseline during the fold associated with the COVID-19 Delta variant, but this gain disappears in the remaining test periods and reverses in the final fold, where performance drops to $R^2 = -0.434$ compared with -0.280 for the financial-only model.

An additional experiment considered VIX direction (increase versus decrease) as a binary classification task. Here, the combined model achieved a slightly higher AUC than the financial-only model (0.594 versus 0.555). However, the sentiment-only model performed worse than the naive baseline, indicating that the result is not consistent across different evaluation settings.

Overall, the evidence suggests that the predictive value of sentiment identified for realized SPY volatility does not generalize to forecasting the VIX itself. Consequently, VIX-based trading strategies are not investigated further in this thesis.

5.7.6. Summary

Across these five checks, the one-day-ahead volatility result is not an artifact of a small number of extreme observations or of an arbitrarily chosen conditioning variable, but it is also not unconditionally stable: it is concentrated in periods of sudden, news-driven market stress, driven primarily by retail attention (message volume) rather than by the direction or dispersion of sentiment, and it does not extend either to a binary risk-flagging framing or to forecasting market-implied volatility directly. These qualifications shape the interpretation of the thesis’s central finding, developed in the concluding section of this chapter.

5.8. Discussion

The results of this chapter provide little evidence that sentiment extracted from financial social media consistently improves market forecasting. Across all return targets, neither the sentiment-only nor the combined model achieves a meaningful improvement over the financial-only baseline. A similar pattern emerges for market direction. Although the combined model performs better than the financial-only model at the one-day horizon, this improvement does not persist at longer horizons, and the sentiment-only model actually achieves the highest performance on the same task. Taken together, these findings suggest that sentiment does not provide a stable source of complementary information for directional forecasting.

The evidence is more convincing for volatility forecasting, although it is considerably more specific than the original research question might suggest. The improvement is confined to the one-day horizon and is strongest during particular market conditions. The robustness analyses show that it survives the inclusion of the VIX in the financial baseline and cannot be explained by a small number of extreme observations. At the same time, the effect is not universal. It weakens or disappears at longer horizons, does not extend to binary high-volatility classification, and does not transfer to forecasting the VIX itself.

The ablation study further clarifies the source of this improvement. While feature importance initially suggested that measures of sentiment dispersion were driving the result, the ablation analysis identifies message volume as the strongest individual contributor. This points towards retail attention, rather than sentiment polarity itself, as the most informative signal. The remaining sentiment features still improve overall performance when used together, indicating that their contribution is complementary, even if individually weaker.

Overall, the results suggest that information extracted from financial social media is more useful for anticipating short-term changes in market uncertainty than for predicting future market direction or returns. More specifically, the findings indicate that retail attention derived from social-media activity provides regime-dependent information that can improve one-day-ahead volatility forecasting under certain market conditions.

Finally, it is important to interpret this result in the context of the prediction timeline described in Section 5.5.1. All forecasts are generated after both the trading session and the day's social-media activity have concluded. The contribution of sentiment should therefore be understood as improving overnight forecasts for the following trading day, rather than providing a signal that could be acted upon during the same trading session.

6 | Conclusions and Future Work

This thesis set out to answer a single question, approached in two stages: does sentiment extracted from financial social media text carry information that is useful for financial forecasting, once measured with a reliable sentiment signal and evaluated against a genuinely demanding financial baseline? Answering it required first building and validating that sentiment signal (Part 1), and then testing it directly against the toughest available market benchmark (Part 2). Taken together, the two stages point to a single, more specific answer than the question first suggests: financial social media do not provide a robust directional trading signal, but retail attention extracted from sentiment-labelled posts contains regime-dependent information for short-term volatility forecasting. This chapter summarises what was learned at each stage, returns explicitly to the research questions posed in Chapter 2, discusses the limitations of the approach taken, and outlines directions for future work.

6.1. Summary of Findings

6.1.1. Part 1: Extracting a Reliable Sentiment Signal

Four increasingly sophisticated approaches were developed to classify StockTwits posts about SPY as *Bullish* or *Bearish*. The overwhelming majority of the achievable improvement came from a single step: adapting a general-purpose financial sentiment model to the target data at all. Fine-tuning FinBERT on labelled StockTwits posts (Approach 2) improved accuracy by more than 22 percentage points over applying the same model zero-shot (Approach 1, Section 4.2), confirming that the distribution shift between FinBERT’s original fine-tuning corpus and short, informal financial social media text is substantial, and that a comparatively simple fine-tuning step closes most of it. The two semi-supervised extensions built on top of this fine-tuned model – self-training with joint confidence-and-neighbourhood pseudo-label filtering (Approach 3), and label spreading over a similarity graph (Approach 4) – each delivered a further, genuine, but small improvement (macro F1 gains of 0.0075 and 0.0089 respectively). This result was tested rather than simply accepted: the assumption that more confident pseudo-labels

are also more often correct was checked directly against withheld ground truth for both approaches; both were retrained at stricter, quality-optimised thresholds to test whether higher label quality translates into a better final classifier; and the possibility that a large labelled set was masking a bigger semi-supervised benefit was ruled out by repeating all three supervised and semi-supervised approaches on shrinking fractions of the labelled data, where the same small gap persisted. On this evidence, Approach 2 – the simpler, purely supervised model – was retained as the sentiment classifier carried into the forecasting stage (Section 4.6), on parsimony grounds rather than because the semi-supervised approaches failed: they constitute, to the best of this thesis’s knowledge, one of the few head-to-head comparisons of these two semi-supervised paradigms on financial social media text under a matched embedding space and evaluation protocol, and are reported in full as a methodological contribution in their own right.

6.1.2. Part 2: Does Sentiment Add Forecasting Value?

The sentiment classifier from Part 1 was used to construct daily sentiment features – level, dispersion, and volume-based, in line with the distinction drawn in Section 3.4 – and combined with a broad set of purely financial features, including the CBOE Volatility Index (VIX), across nine forecasting targets: return, direction, and realised volatility, each at a 1-day, 3-day, and 1-week horizon. Three model families (financial-only, sentiment-only, and combined) were each evaluated with three algorithms and a naive baseline, and compared identically across all nine targets. The result is narrow but clear, and it centres on one outcome in particular: realised volatility, not the direction or level of returns. A model combining sentiment and financial features improves 1-day-ahead volatility forecasting substantially (R^2 improving from 0.048 to 0.125), and this is the result treated as the central, defensible finding of this thesis. Two smaller effects also appear elsewhere among the nine target-horizon combinations tested – a marginal gain for 1-week-ahead returns (R^2 improving from -0.318 to -0.311 , though neither the financial-only nor the combined model beats the naive baseline at that horizon, so returns remain close to unpredictable there regardless of feature set) and a small gain for 1-day-ahead direction (AUC improving from 0.504 to 0.548) – but neither is treated as part of the thesis’s core claim: both are small in absolute terms, and for direction specifically a sentiment-only model outperforms the combined one, indicating a fragile pattern rather than a robust, decision-relevant one. No improvement of any kind is observed at the 3-day or 1-week direction horizons, nor for 1-day or 3-day returns, nor for 3-day or 1-week volatility. The defensible conclusion is accordingly narrower than a claim about sentiment in general: financial social media sentiment does not add forecasting value for the direction

or level of returns, and it does not do so primarily by tracking bullishness or bearishness at all. What it adds instead is a measure of how much the market is talking, and volatility responds to that. This volatility result is directly relevant to the two gaps identified in the literature review (Section 3.5): it holds even though the financial baseline already includes the VIX, the market’s own forward-looking volatility estimate, showing that a text-based signal can add information beyond what a large, liquid options market already prices in; and, once level, dispersion, and volume-based sentiment features are examined individually, it is daily message volume – a direct proxy for retail attention – that carries most of this value, rather than the direction or dispersion of sentiment expressed in those messages. Feature-importance analysis of the best-performing model is consistent with this: the VIX level and trading volume together account for roughly half of total feature importance, and among the sentiment features, message volume contributes more than either the level or the dispersion of sentiment. In other words, the market appears to react less to what retail investors on StockTwits are saying about SPY, and more to how many of them are saying anything at all. A further, exploratory analysis splits trading days into quartiles by two proxies for the same underlying construct – how many posts were made (attention) and how divided sentiment is (disagreement) – and compares the combined and financial-only models within each group on the one result that held up, 1-day volatility. The combined model helps most on the quietest days, with the fewest posts, precisely where the financial-only model performs worst on its own ($R^2 = -0.53$), so that sentiment appears to fill a gap left open by financial data specifically when attention is otherwise low; a related, smaller benefit also appears on the most divided days. This subgroup analysis covers only around 62 trading days per group and should be read as suggestive rather than conclusive, but it points to a coherent and interpretable story: sentiment adds the most value not when the crowd is uniformly optimistic or pessimistic, but when engagement with the stock is unusually low or unusually fractured – that is, when retail attention itself, rather than its direction, is the informative quantity.

6.2. Answering the Research Questions

Returning to the two research questions posed in Chapter 2: first, is a more carefully filtered or more structurally informed pseudo-labelling strategy worth its added complexity relative to a simpler supervised baseline, once labelled financial social media data is genuinely scarce? The evidence here says no, at least at the scale and label quality available in this dataset: both semi-supervised strategies improve on supervised fine-tuning only marginally, and this gap does not widen when labelled data is artificially reduced, suggesting the limiting factor is the intrinsic difficulty of the classification task rather than the

amount of labelled or pseudo-labelled data used to approach it. Second, does a sentiment signal derived from that classifier add value once evaluated against the toughest available financial baseline, and does the answer depend on which construct and which outcome are considered? Here the evidence says yes, but narrowly and specifically: sentiment adds robust forecasting value for short-horizon realised volatility; this value is carried chiefly by daily message volume – a proxy for how much retail attention a stock is receiving – rather than by the direction or dispersion of the sentiment expressed; and the improvement survives comparison against a baseline that already includes the market’s own volatility benchmark. A small same-day effect is also observed for return direction, but it is not part of this defensible conclusion: it does not hold at longer horizons, and at the one-day horizon itself a sentiment-only model outperforms the combined one, indicating the effect is not robustly complementary to financial information in the way the volatility result is. For the level of returns, no feature set improves meaningfully on a naive benchmark at any horizon tested. The value of financial social media, in other words, lies chiefly in what its volume reveals about retail attention – and it does so in a regime-dependent way, concentrated in periods when engagement is unusually low or sentiment is unusually divided – rather than in a reliable directional signal about where prices are headed. Financial social media do not provide a robust directional trading signal, but retail attention extracted from sentiment-labelled posts contains regime-dependent information for short-term volatility forecasting.

6.3. Limitations

Several limitations qualify these conclusions and should be kept in mind when interpreting them. The analysis is based on a single ticker (SPY) and a single social media platform (StockTwits); the extent to which the specific findings – particularly the attention-driven volatility result and the quiet-days effect – generalise to other tickers, asset classes, or platforms is untested. The sample period, March 2020 to April 2022, is dominated by the COVID-19 market crash and its aftermath, an unusually volatile and unusual regime; results obtained here may not transfer directly to calmer markets. The ground truth itself is a self-reported, binary Bullish/Bearish tag rather than an expert or verified label, is available for well under half of all posts, and by construction excludes a genuine neutral category, all of which cap the achievable classification performance regardless of modelling choice, as reflected in the roughly 0.80–0.81 F1 ceiling reached by every fine-tuned and semi-supervised approach in this thesis. The semi-supervised comparison in Part 1, though thorough within its own scope, used a single seed encoder (fine-tuned FinBERT); newer or larger pretrained language models might shift the balance between

supervised and semi-supervised performance in ways this thesis cannot rule out. The forecasting stage relies on classical and tree-based learners (regularised linear models, random forests, gradient boosting) applied to engineered tabular features rather than on sequence models operating directly on the time series of sentiment and price, and may therefore not fully exploit temporal dependencies in the data. The comparison protocol itself is deliberately generous: for each target and feature set, the best-performing of three or four candidate algorithms is reported, in order to show each feature set’s best achievable performance rather than a single fixed pipeline’s performance; this is a reasonable choice for an exploratory comparison, but it is an additional source of optimism in the reported numbers, on top of the nine targets and three feature-set families already being compared. Finally, the subgroup analysis by attention and sentiment dispersion is based on a small number of trading days per quartile and should be treated as an illustrative, hypothesis-generating result rather than a statistically conclusive one. Taken together, these two multiple-comparison sources are the reason this thesis treats only the 1-day volatility result, corroborated independently by feature importance and by the quartile analysis, as its central, defensible finding, rather than the smaller and less consistent effects observed for direction and for 1-week returns.

6.4. Future Work

Several directions follow naturally from these limitations. Extending the analysis to additional tickers and asset classes, and to sample periods covering calmer market regimes, would test whether the volatility result and the attention-driven interpretation found here are a general property of social-media-derived sentiment or specific to SPY during an unusually volatile window. Repeating the semi-supervised comparison in Part 1 with a stronger or more recent pretrained language model as the seed encoder would clarify whether the marginal gains observed here reflect a genuine ceiling in the data, or a ceiling specific to the particular encoder used. On the forecasting side, replacing or complementing the tabular feature approach with sequence models applied directly to sentiment and price time series could capture temporal dynamics – such as the build-up and resolution of attention and disagreement over several days – that the current daily, aggregated features may only partially represent. The finding that sentiment helps most on the quietest and the most divided trading days is itself a natural candidate for a dedicated, purpose-built study, using regime-conditional models estimated directly on attention and disagreement rather than recovered post hoc through a quartile split. Finally, explicitly modelling the uncertainty in self-reported StockTwits labels – for instance, treating them as noisy rather than ground-truth signals during fine-tuning – may offer a more principled route past the

classification ceiling observed throughout Part 1 than either additional labelled data or more elaborate pseudo-labelling would.

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